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Review

V-ATPase Dysfunction in the Brain: Genetic Insights and Therapeutic Opportunities

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Abstract: Vacuolar-type ATPase (v-ATPase) is a multimeric protein complex that regulates H^+ transport across membranes and intra-cellular organelle acidification. Catabolic processes, such as endocytic degradation and autophagy, strictly rely on v-ATPase-dependent luminal acidification in lysosomes. The v-ATPase complex is expressed at high levels in the brain and its impairment triggers neuronal dysfunction and neurodegeneration. Due to their post-mitotic nature and highly specialized function and morphology, neurons display a unique vulnerability to lysosomal dyshomeostasis. Alterations in genes encoding subunits composing v-ATPase or v-ATPase-related proteins impair brain development and synaptic function in animal models and underlie genetic diseases in humans, such as encephalopathies, epilepsy, as well as neurodevelopmental, and degenerative disorders. This review presents the genetic and functional evidence linking v-ATPase subunits and accessory proteins to various brain disorders, from early-onset developmental epileptic encephalopathy to neurodegenerative diseases. We highlight the latest emerging therapeutic strategies aimed at mitigating lysosomal defects associated with v-ATPase dysfunction.

Keywords: v-ATPse; lysosomal dysfunction; neurodevelopmental disorders; neurodegeneration

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Citation: Falace, A.; Volpedo, G.; Scala, M.; Zara, F.; Striano, P.; Fassio, A. V-ATPase Dysfunction in the Brain: Genetic Insights and Therapeutic Opportunities. *Cells* **2024**, *13*, 1441. https://doi.org/10.3390/cells13171441

Academic Editors: Inmaculada Posadas and María Dolores Pérez-Carrión

Received: 6 July 2024
Revised: 23 August 2024
Accepted: 25 August 2024
Published: 28 August 2024

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1. Introduction: v-ATPase Structure and Function

Rotary adenosine triphosphatases (ATPases) play key roles in energy conservation, transmembrane transport, acidification of intracellular compartments, and pH homeostasis. They are divided into three classes based on their structure and the type of ion they transport [F-ATPases (e.g., mitochondrial H^+-ATPase), P-ATPases (e.g., Ca^{2+}-ATPase, Na^+, K^+-ATPase), and V-ATPases (e.g., lysosomal H^+-ATPase)]. Upon ATP hydrolysis, vesicular or vacuolar-type ATPases (v-ATPases) mediate hydrogen ion transport. In mammals, two different domains operating with a rotary mechanism make up this highly conserved multimeric complex: the V1 domain, composed of eight cytosolic subunits, binds and hydrolyses ATP in the cytosol, while the V0 domain, composed of seven transmembrane subunits, forms the hydrogen pore [1] (Figure 1). Upon ATP hydrolysis, V1 drives the movement of a central rotary complex, which results in proton translocation through the V0 domain. V0 and V1 can reversibly associate and dissociate to regulate the activity of the proton pump, which can be modified by several factors, including nutrient availability, hydrogen ion concentration, growth factors, and kinases [2–4]. Dissociation or failed assembly of the V1 and V0 domains hinder ATP-driven proton pumping, leading to a

lack of proton transport across membranes. Maintaining and regulating the pH across the membrane is important in various cellular processes, such as membrane trafficking, protein degradation, autophagy, and small molecule transport [1,5]. In addition, v-ATPases are molecular hubs for sensing and/or transmitting signals through the mTOR, WNT, and NOTCH pathways [6]. Upon binding with the v-ATPase, mechanistic target of rapamycin complex 1 (mTORC1) localizes to the lysosomal surface and promotes the mTOR pathway via the Ragulator–Rag protein complex, thus modulating transcription factor EB (TFEB)-dependent gene cascade [7–12].

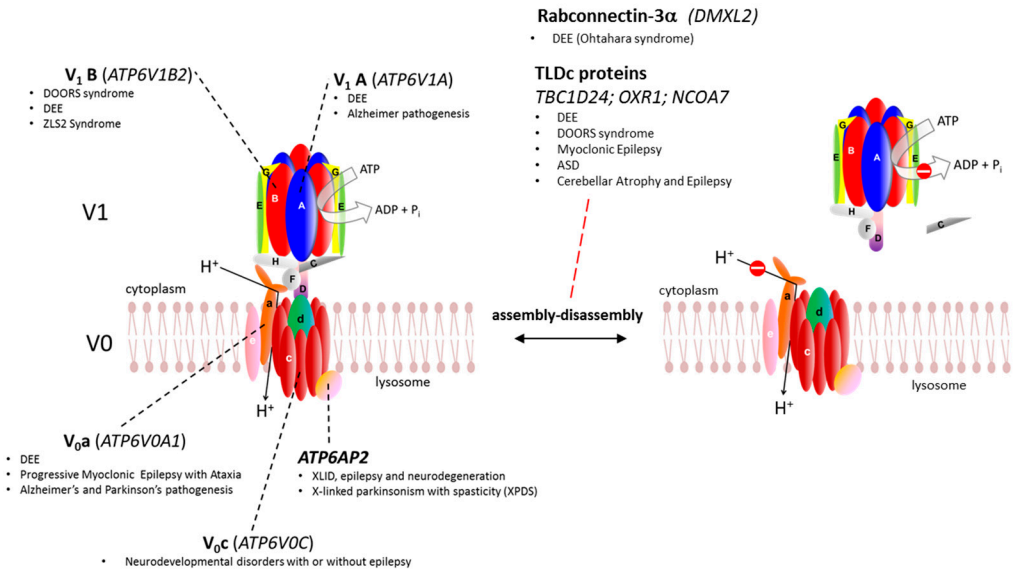


Figure 1. v-ATPase assembly/disassembly. Genetic defects and associated brain diseases are marked for the subunits, accessory protein, and assembly/disassembly regulators. **Legend**: ATP6AP2, ATPase H-transporting lysosomal accessory protein 2; ASD, autism spectrum disorder; DEE, developmental and epileptic encephalopathy; DOORS, deafness, onychodystrophy, osteo-dystrophy, retardation and seizures; XLID, X-linked intellectual disability; ZLS2, Zimmermann–Laband syndrome 2.

In humans, a redundant set of subunits is encoded by 22 autosomal genes, enabling the assembly of distinct v-ATPase complexes composed of subunits expressed in a tissue-specific manner, ultimately resulting in different degrees of regulated acidification [13,14]. The selective expression of specific accessory proteins, such as ATPase H-transporting lysosomal accessory protein 2 (ATP6AP2), is important for directing v-ATPase to distinct tissue types and specific membranes [15].

Due to its key role in cellular homeostasis, pathogenetic variants in v-ATPases-related genes are associated with various disorders spanning from osteoporosis to renal tubular acidosis and neurological diseases [5].

The v-ATPase complex is expressed at high levels in neurons, where it plays additional and unique roles, such as neurotransmitter loading into synaptic vesicles [16,17]. In this context, alterations in genes encoding v-ATPase subunits or their regulators have recently been described in different neurological disorders and in aging.

Here we discuss genetic and functional evidence linking v-ATPase subunits and the most relevant v-ATPase accessory proteins in the pathogenesis of different brain disorders, spanning from early-onset developmental epileptic encephalopathy to neurodegenerative conditions (Figure 1 and Table 1). The most recent emerging findings on therapeutical strategies for v-ATPase-related brain disorders are also discussed.

Table 1. Synopsis of brain disorders related to genetic v-ATPase dysfunction. Abbreviations: AD, autosomal dominant; AR, autosomal recessive; XLR, X-linked recessive; DEE, developmental and epileptic encephalopathy; DOORS, deafness, onychodystrophy, osteodystrophy, retardation and seizures; ZLS2, Zimmermann–Laband syndrome 2; NDD, neurodevelopmental disorder; EEO3, early-onset epilepsy 3; XLID, X-linked intellectual disability; XPDS, X-linked parkinsonism with spasticity; FIME, familial infantile myoclonic epilepsy; EPRPDC.; epilepsy rolandic with paroxysmal exercise-induced dystonia and writer's cramp; CHEGDD, cerebellar hypoplasia/atrophy, epilepsy, and global developmental delay; ASD, autism spectrum disorder.

Human Gene (NCBI Ref.Seq)	Gene Product	Inheritance	Associated Neurogenetic Conditions (OMIM)	Main and Clinical Manifestations
ATP6V1A (NM_001690.4)	V1 subunit A	AD	DEE 93 (OMIM #618012)	delayed psychomotor development; early-onset refractory seizures; impaired intellectual development; cerebral and cerebellar atrophy; progressive neurodegeneration
ATP6V1B2 (NM_001693.3)	V1 subunit B	AD	DOORS syndrome (OMIM #220050); ZLS2 syndrome (OMIM #616455); DEE.; Autosomal dominant epilepsy with or without intellectual disability	delayed psychomotor development; early-onset refractory seizures; impaired intellectual development; cerebral and cerebellar atrophy; progressive neurodegeneration; specific dysmorphic features (DOORS patients)
ATP6V0A1 (NM_001130020.3)	V0 subunit A	AD/AR	DEE 104 (OMIM #619970); NDDs with epilepsy and brain atrophy (OMIM #619971)	delayed psychomotor development; early-onset refractory seizures; impaired intellectual development; cerebral and cerebellar atrophy; progressive myoclonus epilepsy with ataxia
ATP6V0C (NM_001694.4)	V0 subunit C	AD	EEO3, with or without developmental delay (OMIM #620465)	delayed psychomotor development; early-onset refractory seizures; impaired intellectual development; gait ataxia, nonspecific dysmorphic features
ATP6AP2 (NM_005765.3)	ATPase, H+ transporting lysosomal accessory protein 2	XLR	XLID, epilepsy and neurodegeneration (OMIM #300423); XPDS (OMIM #300911)	parkinsonian features and spasticity; delayed psychomotor development; early-onset refractory seizures; impaired intellectual development; cerebral and cerebellar atrophy; progressive neurodegeneration
DMXL2 (NM_001174116.3)	Rabconnectin-3α	AR	DEE 81 (OMIM #618663)	delayed psychomotor development; early-onset refractory seizures; impaired intellectual development; cerebral and cerebellar atrophy; progressive neurodegeneration; facial dysmorphism; progressive leukoencephalopathy

Table 1. *Cont.*

Human Gene (NCBI Ref.Seq)	Gene Product	Inheritance	Associated Neurogenetic Conditions (OMIM)	Main and Clinical Manifestations
TBC1D24 (NM_001199107.2)	TBC1 Domain Family Member 24	AR	FIME (OMIM #605021); DEE 16 (OMIM #615338); DOORS syndrome (OMIM# 220050); EPRPDC (OMIM# 608105)	delayed psychomotor development; early-onset refractory seizures; impaired intellectual development; myoclonic epilepsy (FIME patients), specific dysmorphic features (DOORS patients); exercise-induced dystonia (EPRPDC patients)
OXR1 (NM_001198532.1)	Oxidation Resistance 1	AR	CHEGDD (OMIM# 213000)	delayed psychomotor development; impaired intellectual development; seizures; cerebellar atrophy and dysplasia; ataxia; nonspecific dysmorphic features
NCOA7 (NM_001199619.2)	Nuclear receptor coactivator 7	AR	ASD	autistic features

2. v-ATPase Gene Variants in Brain Disorders

2.1. Genes Encoding Subunits of the V1 Domain

The V1 cytosolic domain, composed of eight distinct subunits (A–H), has the function of binding and hydrolyzing ATP [1] (Figure 1). Genetic mutations of this domain can compromise enzymatic activity, as well as impairing subunit interactions and pump rotation. The core of the V1 domain is composed of a protein ring with three couples of alternate V1A and V1B subunits (Figure 1). Altered function of the V1 domain in the brain leads to altered pH homeostasis and ultimately to the development of neurological disorders as consequence of autophagic failure and synaptic defects. In particular, mutations in *ATP6V1A*, encoding V1A subunit, can promote developmental and epileptic encephalopathy (DEE) and its downregulation was associated with late-onset Alzheimer's disease (AD), while mutations in *ATP6V1B2*, encoding the V1B subunit, mediate DEE, deafness, onychodystrophy, osteodystrophy, intellectual disability, and seizure (DOORS) syndrome, and Zimmermann Laband Syndrome (ZLS) (Figure 1).

2.2. De Novo Mutations of the ATP6V1A Gene Cause DEE

Recessive mutations in the *ATP6V1A* gene, coding for the V1A subunit, were first identified in patients presenting with cutis laxa, dysmorphic features, and seizures [18]. More recently, several de novo missense mutations in *ATP6V1A* have been described in patients with DEE of variable severity, with symptoms ranging from moderate intellectual disability (ID) and seizures to early-onset DEE with premature lethality [19,20]. The affected individuals (age range: 6 weeks–22 years old) primarily showed a spectrum of neurodevelopmental phenotypes of different degrees of severity, including small head size, hypotonia, ID, multiple seizure types (mostly infantile spasms), and motor disorders. At a mean age of 7 years old, the clinical picture included early lethal encephalopathies with rapidly progressive and massive brain atrophy, severe DEE, and ID with epilepsy. Less severe outcomes (25% incidence) were represented by moderate ID and variable epilepsy. Additionally, less than half of the patients showed microcephaly and amelogenesis imperfecta/enamel dysplasia. Brain MRIs demonstrated hypomyelination, as well as generalized and progressive atrophy in two-thirds of the patient cohort [20].

Additional evidence for the deleterious effects of *ATP6V1A* variants was provided by structural modelling, thanks to the recently solved atomic structure of the mammalian brain v-ATPase [21,22]. Thirteen of the *ATP6V1A* mutations described in this study were located

on ATP-binding or catalytic sites at the interface of subunits A and B, indicating that they may directly impair enzymatic activity. Modelling of the remaining variants suggests that some can hinder v-ATPase rotation or subunit interactions, while others may alter protein expression and stability [20]. Taken together, these observations highlight the importance of the V1 subunit A, and ultimately of a functional v-ATPase, for correct neurodevelopment.

In vitro functional experiments further consolidated the pathogenetic role of *ATP6V1A* variants and are consistent with lysosomal impairment and loss of v-ATPase function in patients' cells. Fibroblast cultures from two *ATP6V1A*-related DEE patients showed decreased expression of the lysosomal marker LAMP1 and LysoTracker staining, as well as increased organelle pH. In contrast, patients' cells with milder symptoms exhibited an opposite phenotype, suggesting that v-ATPase 'gain-of-function' may elicit milder phenotypes [20]. Ultrastructural analysis in both patients' derived fibroblasts and induced pluripotent cell-derived neurons revealed ultrastructural defects in autolysosomes, which displayed accumulation of non-degraded material of largely unknown composition [20]. When expressed in postsynaptic rodent neurons, *ATP6V1A* pathogenetic variants impaired neurite development and excitatory synaptic connections [19]. Additional heterozygous variants have been described in patients with favorable epilepsy resulting in ATP6V1A loss of expression [23,24].

2.3. ATP6V1A and Neurodegeneration

A multi-omic molecular analysis of late-onset Alzheimer disease (LOAD) across four brain regions in 364 donors with varying cognitive and neuropathological phenotypes identified *ATP6V1A* as a top key regulator of the most dysregulated neuronal subnetwork in LOAD [25,26]. Furthermore, *ATP6V1A* was consistently downregulated in multiple brain regions in individuals with dementia, including those in the early stages of the disease [27]. The *ATP6V1A* gene was also identified as a disease driver in a parallel study characterizing the molecular heterogeneity of Alzheimer disease (AD), by using an integrative network approach on 1543 transcriptomes across five brain regions in two AD cohorts [25].

Experimental in vitro approaches demonstrated that CRISPR/Cas9-mediated *ATP6V1A* downregulation in human iPSC-derived excitatory neurons impairs electrical activity and promotes a selective loss of the synaptic vesicle markers SYN1 and vGLUT1, suggestive of pre-synaptic dysfunction. Neuronal excitability was further compromised by the presence of the toxic amyloid-beta (Aß42) peptide, which has a well-known association with AD pathogenesis [25]. In vivo studies in a Drosophila model genetically ablated for the two orthologs of *ATP6V1A* showed that reduced neuronal expression was sufficient to impair motor function. These changes were accompanied by diminished mRNA levels of synaptic genes, corroborating the synaptic dysfunction found in hiPSC-derived neurons. Aß42 expression alone was also associated with reduced expression of the *ATP6V1A* orthologs in the fly brain [25]. Furthermore, in a recent study, Esposito et al. showed that *Atp6v1a* silencing in rat hippocampal neurons affected synaptogenesis and plasticity, as well as lysosomal and autophagic function [28]. Taken together, these pre-clinical and clinical studies highlight the role of the *ATP6V1A* gene as a driver for synaptic dysfunction and subsequent neuronal loss, and a potential therapeutic target to prevent neurodegeneration.

2.4. ATP6V1B2 De Novo Truncating Variant in DOORS Syndrome

Loss of *ATP6V1B2*, a gene coding for the B2 subunit of the v-ATPase V1 cytosolic domain, is non-compatible with life. In fact, complete deletion of *ATP6V1B2* results in embryonic lethality prior to organogenesis in mice (https://www.mousephenotype.org/data/genes/MGI:109618 (accessed on 5 July 2024). In humans, a recurrent truncating variant in *ATP6V1B2* has been associated with DOORS syndrome. Beauregard-Lacroix et al. reported the *ATP6V1B2* variant c.1516C>T.; p.Arg506* in nine individuals from eight unrelated families with DOORS syndrome [29]. To characterize the function of the *ATP6V1B2* p.R506* variant in brain circuits, a knock-in *Atp6v1b2*emR506* mouse model was generated by CRISPR/Cas9-mediated gene editing [30]. No spontaneous seizures were

recorded in homozygous *Atp6v1b2*emR506* mice; however, continuous intracranial video-EEG recordings revealed that P90-P110 *Atp6v1b2*emR506* mutant mice presented frequent interictal epileptic activity and stimuli-induced seizures. In addition, *Atp6v1b2*emR506* mice displayed seizure susceptibility to a range of pro-convulsive (Pentylenetetrazole) triggers [30], highlighting the role of this subunit in proper v-ATPase functioning and brain homeostasis.

2.5. ATP6V1B2 Recurrent De Novo Missense Variants in Zimmermann-Laband Syndrome

Zimmermann–Laband syndrome (ZLS) is a developmental disorder characterized by facial dysmorphism with gingival enlargement, ID, hypoplasia or aplasia of nails and terminal phalanges, as well as hypertrichosis. In 2015, Kortum et al. identified a recurrent missense change in *ATPV1B2* (NM_001693.3 c.1454G-C, p.Arg485Pro) in two unrelated ZLS patients [31]. Bioinformatic analysis showed that the arginine-to-proline substitution in ATP6V1B2 was predicted to perturb inter-subunit interactions within the V1 subcomplex by destabilizing the C-terminal segment of the B subunit, which is part of the A-B interface and interacts with the E subunit [31]. Destabilization of the V1 domain leads to v-ATPase dysfunction, highlighting the crucial role for appropriate pH regulation and v-ATPase-mediated signaling during brain development.

2.6. ATP6V1B2 Pathogenetic Variants in Epileptic Syndromes and DEE

In 2017, Popp et al., reported an heterozygous change (c.1120G>C, p.Glu374Gln) in *ATPV1B2* in an individual with severe ID, hypotonia, microcephaly, and seizures [32]. Subsequently, Shaw et al. described a large Polish family with relatively mild gingival and nail problems, no phalangeal hypoplasia, and with generalized epilepsy, harboring a novel dominant missense variant (c.1192C>G, p.Leu398Val) in *ATP6V1B2* [33]. More recently, de novo variants in *ATP6V1B2* have also been found in DEE patients. Inuzuka et al. reported the case of an infant with severe epileptic encephalopathy, microcephaly, and profound developmental delay associated with a novel de novo loss-of-function variant (c.1465A>T, p.Lys489*) in *ATP6V1B2*, diagnosed by whole-exome sequencing [34]. Furthermore, exome sequencing identified a novel de novo variant in *ATP6V1B2* (c.973G>C, p.Gly325Arg) in a patient affected by global developmental delay, skeletal abnormalities, and epileptic encephalopathy featuring Lennox–Gastaut syndrome [35]. Taken together, these data identify a role for the V1B subunit in neuronal excitability and seizures.

2.7. Genes Encoding Subunits of the V0 Domain

The V0 transmembrane domain forms the hydrogen pore and consists of seven distinct subunits [1] (Figure 1). Variants in the V0A subunit can promote DEE and myoclonic epilepsy with ataxia, while V0C variants lead to neurodevelopmental disorders with or without epilepsy (Figure 1). Moreover, defects in V0A subunit function drive physiopathological events in early-onset AD.

2.8. De Novo and Biallelic ATP6V0A1 Variants Cause DEE and Progressive Myoclonic Epilepsy with Ataxia

ATP6V0A1 encodes for the brain-enriched isoform of the A subunit and is part of the CLEAR (coordinated lysosomal expression and regulation) network of genes regulated by TFEB [36]. Recently, two studies have reported various de novo and biallelic *ATP6V0A1* variants as major causes of DEE [37,38]. Biallelic variants in this gene have been identified also in five patients affected by early-onset progressive myoclonus epilepsy with ataxia [38]. Functional studies performed in the *Caenorhabditis elegans* model showed that the recurrent DEE pathogenetic variant p.Arg740Glu leads to failure of lysosomal activity by directly impairing acidification of the endo-lysosomal compartment, thus causing autophagic dysfunction and severe developmental defects [38]. Neuro2a cell lines stably expressing the p.Arg740Glu variant displayed impaired protonation and decreased level of the lysosomal enzyme Cathepsin D, accompanied by the accumulation of the autophagosomal marker

LC3-II [38]. Moreover, Aoto et al. developed a knock-in mouse model harboring either the human p.Arg740Glu or the p.Ala505Pro variant, also associated with DEE [37]. *Atp6v0a1* R740Q/R740Q mice do not survive the embryonic stage, suggesting that this mutation leads to a complete v-ATPase loss-of-function, which is not compatible with life. *Atp6v0a1* A505P/A505P mice survived for 2 weeks and displayed impaired motor function and ataxia [37,39]. *Atp6v0a1* A505P/A505P brain staining revealed reduced numbers of layer V neurons and impaired synaptogenesis. Furthermore, the decreased levels of the Atp6v1a protein in the *Atp6v0a1* A505P/A505P model suggested that the Ala505Pro variant affects the stability of the v-ATPase complex [37]. Investigations of lysosomal and autophagic function revealed increased neuronal cell death, abnormal cellular distribution of lysosomes, decreased cathepsin D activity, as well as autophagosome and lysosome accumulation in the *Atp6v0a1* A505P/A505P mouse brain [37]. These phenotypes were accompanied by a concomitant decrease of mTORC1 activity. Furthermore, electrophysiological recordings in *Atp6v0a1* A505P/A505P cultured neurons demonstrated a reduction in the amplitude and frequency of both miniature excitatory and inhibitory postsynaptic currents, suggestive of lowered neurotransmitter content in the synaptic vesicles and decreased neurotransmitter release [37,39]. Taken together, these studies highlight the pathogenic role of *ATP6V0A1* variants during neurodevelopment, impacting both the number and function of different neuronal populations.

2.9. ATP6V0A1 and Neurodegeneration

Pathogenetic variants in *Presenilin 1* (*PS1*) are the most common cause of early-onset familial AD (FAD) [40]. *PS1* encodes for a ubiquitous transmembrane protein showing complex roles in cell adhesion, apoptosis, neurite outgrowth, calcium homeostasis, and synaptic plasticity [41]. Following cleavage at the endoplasmic reticulum, PS1 participates in the gamma secretase complex, which mediates the intramembranous cleavage of amyloid-beta precursor proteins (APPs) [41]. Disease-causing *PS1* mutations increase the generation of the neurotoxic amyloid-β peptide from APP, a classical hallmark of AD. PS1 facilitates the maturation and delivery of the v-ATPase V0a1 subunit to the lysosome [41]. Therefore, lysosomal acidification and autophagy are severely impaired in the brain of *PS1*-deficient mice. Of note, autophagic failure and v-ATPase mistargeting were also observed in fibroblasts from patients with FAD harboring deleterious variants in *PS1* [41]. Constitutively, phosphorylated APP selectively interacts with the cytoplasmic domain of the V0a1 subunit, thus preventing association of the V0/V1 complex in vitro and in vivo [42]. v-ATPase lysosomal acidification was defective and v-ATPase assembly disrupted in fibroblasts from patients with Down syndrome (DS) [42], the congenital genetic disorder responsible for the most prevalent form of early-onset AD [43]. These findings were also confirmed in the brain tissue from a DS animal model (*Ts2* mouse) and from a DS patient [41]. Reducing APP phosphorylation can rescue v-ATPase function and lysosome acidification in DS fibroblasts and in the animal model [39]. Lastly, V0a1 subunit dysfunction can lead to lysosomal defects underlying neurodegenerative conditions, such as *LRRK2*-linked PD [44] and *CLN1*-associated disorders [45], highlighting the importance of this subunit in maintaining healthy brain functions.

2.10. De Novo Heterozygote ATP6V0C Variants in Patients with Neurodevelopmental Disorders with or without Epilepsy

ATP6V0C encodes for the 155-amino acid c-subunit of the V0 domain which, along with the c″ subunit (encoded by *ATP6V0B*), forms the intramembrane c-ring that facilitates the movement of protons across the membrane [46]. Mucha et al. described a 16p13.3 microdeletion resulting in simultaneous haploinsufficiency of *TBC1D24*, *ATP6V0C*, and *PDPK1*, underlying a neurological syndrome characterized by microcephaly, developmental delay, intellectual disability, and epilepsy [47]. Interestingly, despite the presence of the epilepsy gene *TBC1D24* in the genomic deletion, *ATP6V0C* haploinsufficiency was proposed as the primary contributor to the clinical features of the 16p13.3 microdeletion syn-

drome [48]. The identification of a de novo stop-loss *ATP6V0C* variant in an individual with epilepsy and ID further corroborated this hypothesis [49–51]. More recently, Mattison et al. identified heterozygous *ATP6V0C* missense variants in 27 patients with a novel syndrome of developmental delay, epilepsy, and ID [52] (OMIM# 620465). Generalized tonic–clonic, focal, atonic, and myoclonic seizures were reported in 19 patients of the cohort with a mean age onset of 24.6 ± 8.0 months. Common MRI findings included agenesis/hypoplasia of the corpus callosum or cerebellar vermis and delayed myelination. In silico modelling suggested that the patient variants interfered with the interactions between the *ATP6V0C* and *ATP6V0A* subunits during ATP hydrolysis [52]. The deleterious effects of *ATP6V0C* variants were also assessed by functional analyses in *Saccharomyces cerevisiae* and in invertebrate models. Three selected variants modelled in *Caenorhabditis elegans* led to motor dysfunction, as well as reduced growth and survival [52]. Knockdown of *Dmel\Vha16–3*, the *ATP6V0C* orthologous in *Drosophila*, resulted in seizure-like behavior and pretreatment of these larvae with a variety of established antiepileptic drugs resulted in significant reductions in recovery time [52], consistent with the hypothesis that haploinsufficiency of *ATP6V0C* contributes to seizures in multiple disease models.

3. The Pathogenetic Role of V-ATPase Accessory Proteins

v-ATPase is accompanied by accessory proteins which can enhance stability and regulate the function of the pump. The ATP6AP2 accessory protein is associated with the transmembrane domain and plays a role in v-ATPase assembly and lysosome acidification. Furthermore, DMXL2 plays a role in V0/V1 assembly and signal transduction. Mutations in their encoding genes can lead to neurodegenerative disorders, such as Parkinson's disease (*ATP6AP2*) and Ohtahara syndrome (*DMXL2*). More recently, the modulatory roles of Tre2/Bub2/Cdc16 (TBC), lysin motif (LysM), and domain catalytic (TLDc) protein family members on v-ATPase function have also emerged. TLDc family genes include *TBC1D24*, *NCOA7*, and *OXR1*, which have been extensively associated with neurological conditions.

3.1. De Novo ATP6AP2 Variant in X-Linked Intellectual Disability (XLID), Epilepsy and Neurodegeneration

The *ATP6AP2* (ATPase, H+ transporting, lysosomal accessory protein 2) gene on Xp11.4 encodes for a v-ATPase accessory protein firstly identified for its role as a (pro)renin receptor in the renin–angiotensin system [53]. *ATP6AP2* is highly expressed in the CNS from the earliest stages, and functional studies performed in yeast and flies suggest that ATP6AP2 participates in the assembly of the v-ATPase proton pore in the endoplasmic reticulum [54]. ATP6AP2 plays a role in protein degradation and regulation of different signaling pathways involved in brain development and synaptic transmission. It is also predicted to enable signaling receptor activity [55,56].

A hemizygous silent mutation in *ATP6AP2* was identified in a putative exonic splicing enhancer site in a kindred of seven males [57,58]. These patients featured global developmental delay apparent from infancy and severe progressive neurologic decline with abnormal movements, spasticity, and seizures. The reported mutation resulted in inefficient inclusion of exon 4 of *ATP6AP2* gene [58]. Subsequently, a hemizygous 2-bp deletion in intron 3 of the *ATP6AP2* was identified in a male infant with a neurodevelopmental disorder and fulminant degeneration [59]. In patient-derived neurons, loss of *ATP6AP2* results in premature and/or ectopic differentiation, as well as accumulation of abnormal vacuoles and decreased spontaneous activity [59]. These defects were associated with severe deficiency in lysosomal acidification and protein degradation, thus leading to neuronal cell death [59]. Telencephalic *Atp6ap2* conditional knockout male mice showed a postnatal growth delay and died at around 4 weeks of age as a consequence of premature progenitor differentiation and the subsequent fulminant death of newborn neurons [59]. These findings were recapitulated in the fly model, suggesting that a loss of ATP6P2 leads to a lack of v-ATPase activity, impairment of autophagy, and severe neurodegeneration mimicking the

pathogenesis observed in human mutations [59,60]. Altogether, these findings highlight the importance of v-ATPase accessory proteins in maintaining proper brain function.

3.2. Altered Splicing of ATP6AP2 Causes X-Linked Parkinsonism with Spasticity

In affected members of a family with X-linked parkinsonism with spasticity (XPDS), originally reported by Poorkaj et al. and Korvatska et al., identified a pathogenetic splice variant in the *ATP6AP2* gene, resulting in an internal deletion of 32 amino acids in the ATP6AP2 protein [61,62]. Postmortem patient brain tissue showed decreased ATP6AP2 levels in the frontal cortex and striatum, as well as a massive accumulation of p62 in the striatum, suggesting impaired autophagy and a defect in lysosome-mediated protein degradation [61]. In vitro *ATP6AP2* knockdown in a heterologous system led to v-ATPase pump dysfunction and to the accumulation of autophagosomes and autolysosomes [62]. More recently, Edelman et al. demonstrated that such internal deletion resulted in a 50% decrease of full-length ATP6AP2 in neuroprogenitor cells derived from XPDS patients, leading to the downregulation of networks related to neuronal fate commitment and axon guidance [63]. Taken together, these results highlight the sensitivity of brain cells to *ATP6AP2* gene expression levels and propose a link between PD and dysregulation of autophagy and protein degradation, mediated by v-ATPase activity.

3.3. Biallelic DMXL2 Variants Cause Ohtahara Syndrome with Progressive Course

The *DMXL2* gene encodes for rabconnectin-3α, a member of the WD40 repeat (WDR) protein family that is highly expressed in the brain. Rabconnectin-3α interacts with the synaptic vesicle (SV)-associated G-protein Rab3A at the synapse and is strictly involved in the assembly and incorporation of v-ATPase into SVs through its Rav1p domain, homologous to the yeast v-ATPase regulator Rav1p [64–67]. Loss of DMXL2 is associated with impaired v-ATPase activity in multiple experimental models [68–71], while biallelic mutations have been found in patients from three unrelated families with severe and rapidly progressing DEE associated with Ohtahara syndrome and premature death [72]. An additional homozygous variant in *DMXL2* has been reported in a patient with consanguineous parents and presenting DEE and macrocephaly [73]. *DMXL2*-related DEE manifested from the first day of life and was associated with deafness, mild peripheral polyneuropathy, and dysmorphic features. Repeated brain MRI investigations from the first months to the first years of life revealed progressive moderate brain shrinkage with leukoencephalopathy [72]. Functional studies performed in *DMXL2* patient fibroblasts showed enhanced LysoTracker signal associated with decreased endo-lysosomal markers and impairment of degradative processes. These defects were accompanied by impaired autophagy, revealed by diminished LC3II signal and accumulation of the autophagy receptor p62, with morphological alterations of the autolysosomal structures [72]. Altered lysosomal homeostasis and defective autophagy were recapitulated in *Dmxl2*-silenced mouse hippocampal neurons, which exhibited impaired neurite elongation and synaptic loss. Complete loss of *Dmxl2* is lethal at the embryonic stage in mice [66,67], whereas heterozygous *DMXL2* mice show macrocephaly and corpus callosum dysplasia [74], uncovering the relevance of early v-ATPase regulation in brain development and function.

3.4. Neurodevelopmental Disorders Associated with TLDc Domains Encoding Genes

Genes encoding TLDc proteins are associated with multiple neurodevelopmental disorders, although the physiopathological bases have not been fully elucidated. Human proteins sharing the TLDc domain include OXR1, NCOA7, TBC1D24, mEAK7, and TLDC2 and have been recently defined as a new class of V-ATPase-associated proteins [75]. Proteomic analyses of v-ATPase-associated proteins revealed that NCOA7 and OXR1 bind to the B1 subunit isoform of the V-ATPase [71]. Subsequent studies extended such interaction to all five members of the TLDc family and to additional cytosolic V1 subunits [76,77]. Castroflorio et al. showed that in primary cortical neurons NCOA7 depletion prevented v-ATPase holoenzyme assembly thus resulting in lysosomal deacidification and autophagic

failure [76]. In contrast, Oot et al. showed that four purified human TLDc proteins (OXR1, NCOA7, TBC1D24, and TLDC2) inhibit V-ATPase by inducing its disassembly in vitro, whereas mEAK7 enhances v-ATPase activity [78]. Thus, the specific role of the different TLDc proteins on the dynamics of v-ATPase assembly/disassembly needs to be determined in a physiological context.

Amongst all TLDc domain-containing genes, *TBC1D24* is the most interesting in the context of this review, due to its role in the pathogenesis of a wide spectrum of neurological conditions, extending from benign idiopathic epilepsies to severe DEEs [79–86]. Interestingly, the clinical phenotype of *TBC1D24* biallelic variants may overlap with that of *ATP6V1B2* mutations, as both genes represent the major genetic cause of DOORS syndrome [87,88], implying a physiological link between TBC1D24 and v-ATPase function. At the cellular level, TBC1D24 impairment has been associated with neuronal phenotypes ranging from early developmental defects to synaptic dysfunction [89–91]. TBC1D24 is required for SVs and receptor trafficking [91–93], suggesting that the interplay between TBC1D24 and v-ATPase may occur at the synaptic compartments. However, further studies are needed to define the precise regulatory role of TBC1D24 in regulating v-ATPase function in the brain.

In addition to *TBC1D24*, two other TLDc domain-containing genes, *NCOA7* and *OXR1*, have been recently associated with neurodevelopmental disorders. The NCOA7 protein interacts specifically with the cytoplasmatic V1 subunits of the v-ATPase and is required for its proper assembly and activity in the brain [76]. A homozygous nonsense mutation in *NCOA7* was characterized in a recessive case of autism spectrum disorder [94]. Neurons lacking NCOA7 exhibit altered development accompanied with defective lysosomal formation and function. The *Ncoa7* deletion animal model exhibited abnormal neuronal patterning defects and a reduced expression of lysosomal markers [76].

Oxidation resistance 1 (OXR1) has emerged as a critical regulator of neuronal survival in response to oxidative stress [95–97]. A recent study showed that the OXR1 protein regulates an alternative pathway of v-ATPase holoenzyme disassembly [98]. This mechanism is ATP-independent and does not require the participation of the RAVE complex [75]. Loss-of-function in the gene's *OXR1* variants was associated with cerebellar atrophy and other severe neurological conditions [99,100]. Consistent with these observations, neuron-specific depletion of *mtd*, the homolog of *NCOA7* and *OXR1* in flies, led to an aberrant lysosomal accumulation, massive neuronal loss, and early death [99]. *OXR1* patient-derived neurons displayed impaired neural differentiation and a dysregulated transcriptome [100]. Taken together, these studies show that aberrant mutations in TLDc proteins lead to impaired v-ATPase assembly and function, and ultimately to a wide spectrum of neurological conditions. However, due to the profound variability in clinical phenotypes, additional studies are needed to tease out the specific mechanisms at play.

4. Therapeutical Potential of V-ATPase Regulation in the Brain

4.1. Direct Modulation of the V-ATPase Activity or Expression

Modulating v-ATPase activity has previously shown therapeutic potential in a variety of disorders, ranging from cancer to viral infections [15]. V-ATPase also plays a role in protection against oxidative stress, a common feature in many neurodegenerative disorders [101]. Thus, developing targeted interventions on tissue-specific v-ATPases can represent a therapeutic option against disorders of different etiology.

To this end, Chung et al. discovered a small-molecule activator of autophagy, EN6, that acts through covalent targeting of a unique regulator to cysteine 277 of the ATP6V1A subunit, thus blocking mTORC1 lysosomal localization and activation [102]. In particular, the EN6-mediated ATP6V1A modification decouples v-ATPase from Rags, leading to inhibition of mTORC1 signaling. Notably, due to its high specificity for mTORC1, unlike other available drugs targeting the mTOR pathway, treatment with EN6 had no significant off-target effects on Akt signaling pathways [102]. Upon EN6 treatment TFEB transcriptional gene targets, including v-ATPase components were significantly increased thus promoting

lysosomal acidification and autophagy progression. EN6 boosts cellular clearance of toxic protein aggregates in inducible Tar-binding protein 43 cells and in vivo treatment with EN6 significantly inhibited mTORC1 signaling in both skeletal muscle and the heart [102].

Direct enhancement of vATPase subunit gene expression represents an alternative and valid approach. For instance, Wang et al. demonstrated that the histone deacetylase inhibitor NCH-51 acts as a selective enhancer of *ATP6V1A* expression, thus normalizing neuronal dysfunction and neurodegeneration caused by loss of ATP6V1A [25]. In human-cultured *ATP6V1A*-silenced neurons, 24 h exposure to 3 μM NCH-51 was adequate to significantly increase ATP6V1A and its related subnetwork thus restoring neural excitability. Administration of 50 μM NCH-51 during aging in a Drosophila model genetically ablated for the two *ATP6V1A* orthologs, dampened Aß42-associated neurodegeneration, and partially increased the mRNA levels of synaptic genes, suggesting that NCH-51 confers neuroprotective effects by correcting neuronal activity [22].

In addition, a role for Rifampicin, an ansamycin antibiotic commonly used against tuberculosis, in the modulation of v-ATPase subunit expression has been proposed. In previous studies, Rifampicin was shown to have a neuro-protective function in acute and chronic brain injury by regulating inflammatory and the PI3K/Akt pathways [103–105]. More recently, Liang et al. showed that rifampicin inhibits rotenone-induced microglia inflammation via improving lysosomal function and autophagy clearance. This effect was mediated by a concomitant increase in the expression of *ATP6V0A1*, encoding the V0 subunit A [106]. Although the precise molecular mechanism linking rifampicin and ATP6V0A1 levels remains unknown, this finding may open an unexplored therapeutical potential for rifampicin in disorders associated with v-ATPase dysfunction.

Natural compounds have also been shown to modulate ATP6V1A activity. FK506 (also kwon as Tacrolimus) is a metabolite produced by the fungus *Streptomyces tsukbaenis*, which acts as strong immunosuppressor by inhibiting the calcium sensitive protein phosphatase calcineurin in T lymphocytes [107]. Kim et al. reported that FK506 selectively bound the V1A subunit and exerted neuroprotective effects by promoting autophagy following activation of the TFEB pathway [108]. Furthermore, dendrobium nobile Lindl. alkaloids (DNLAs), originally extracted from the traditional Chinese herbal medicine, *Dendrobium nobile*, displayed neuroprotective features and promoted autophagy in an animal model of axonal degeneration [109]. Nie et al. showed that DNLA improved lysosome acidification in the hippocampi of amyloid precursor protein/presenilin 1 (*APP/PS1*) mutant mice by increasing the expression of *ATP6V1A* [110].

In contrast with these studies, constitutive downregulation of *ATP6V1A* has been shown to be a potential therapeutical option in amyotrophic lateral sclerosis [111]. Treatment with the FDA-approved ATP6V1A blocker etidronate, a bisphosphonate largely used in osteoporosis treatment [112], decreased levels of the amyotrophic lateral sclerosis associated protein ataxin-2 in cellular models and brains of adult *Ataxin-2* mice [111]. Despite disagreeing with previous data linking lower ATP6V1A levels with pathogenicity [19,20,25,28,113], these results emphasize the importance of ATP6V1A as a therapeutic target for both novel and repositioned drugs, as its regulation is imperative to support neural function and survival. Early intervention on *ATP6V1A* expression and function could prevent neurodegeneration and promote synaptic integrity and plasticity in *ATP6V1A*-mediated disorders.

4.2. Indirect Strategies for Compensating V-ATPase Dysfunction

Contrasting the acidification defects associated with v-ATPase dysfunction may constitute an alternative therapeutical strategy for the here-described neurological disorders. For instance, EMD87580 a benzoyl-guanidine derivative, also known as rimeporide, is a potent inhibitor of the Sodium (Na^+)/proton(H^+) exchangers (NHE), which transport sodium in exchange for proton across lipid bilayers [114]. NHE channel isoforms could reside on the plasma membrane (NHE1–5) to maintain cytoplasmatic pH or at intracellular organelles (NHE6–9), thus regulating luminal acidification [115]. EMD87580, first described

as selective blocker of NHE1 channel [116,117], has been reported as a strong antagonist of the endo-lysosomal NHE6 isoform [118]. Xian et al. showed that blocking NHE-6 with EMD87580 lowered endosomal pH by hindering proton leak in early endosomes. This, in turn, prevented the aggregation of the Alzheimer's disease-associated Apolipoprotein E4 and restored physiological trafficking of ApoE receptor and AMPA- and NMDA-type glutamate receptors [118].

EMD87580 has a good safety and tolerability profile in adults and children and has been selected for different clinical phase I studies, including for congestive heart failure and Duchenne muscular dystrophy (DMD) [116]. Previtali et al. tested EMD87580 with the main goal of preventing or slowing the progression of myocardial damage in DMD children. Pharmacodynamic biomarkers analysis showed encouraging early changes of efficacy and biological effects after 4 weeks of treatment in 20 DMD male patients [116]. Furthermore, 24 h post-stroke EMD87580 treatment in rats, accelerated motor and cognitive function recovery and reduced microglial inflammatory activation and white matter demyelination [119], suggesting the efficacy of EMD87580 treatment also in neurological conditions.

The CIC-7 channel, belonging to the CIC family of chloride (Cl^-) transport membrane proteins may represent an alternative route for contrasting v-ATPase-dependent deacidification. CIC7 is located to late endosomes and lysosomes where it allows the import of Cl^- ions and acts as counterion conductance channel to prevent hyperpolarization of the lysosomal membrane driven by v-ATPase-mediated proton import [120,121]. Lee and colleagues showed that clinically used agent isoproterenol (ISO), belonging to the β2-adrenergic receptor agonist family, normalized elevated pH in *PS1* KO cells and primary fibroblasts derived from *PS1* patients with FAD [122]. ISO is a synthetic catecholamine derived from noradrenaline and was commonly used as a bronchodilator before being dismissed because of its effect on cardiac β receptors [123]. In vitro, 6 h exposure to ISO stimulates lysosomal translocation of ClC-7 from endoplasmic reticulum thus normalizing Cl^- levels and upregulating lysosomal Cl^- channel function in these models. This, in turn, prevented lysosomal hydrolase impairment and autophagic failure restoring lysosomal proteolysis and calcium homeostasis in *PS1* in vitro models [122].

TMEM175 is a lysosomal membrane potassium (K^+) channel, recently described as acting as proton channel at acidic pH and representing a proton leakage pathway thus regulating, in concert with v-ATPase, the maintenance of the correct acidic pH in lysosomal compartments [124]. Interestingly, numerous genetic studies have identified TMEM175 as a risk gene for Parkinson's disease [125–128]. Exposure to the K^+ channel blocker 4-aminopyridine and to high levels of Zn^{2+} have been shown to inhibit TMEM175 activity [129,130]; however, the associated undesirable effects mean that these approaches are not achievable. The design of drugs that selectively act on TMEM175 may represent a further opportunity to treat neurological disorders associated with v-ATPase dysfunction.

The use of nanomaterials as nanoparticles may offer an additional strategy to correct lysosomal pH in a v-ATPase-independent manner. Due to their nontoxic nature and their ability to be internalized by neurons [131], nanoparticles are considered effective and convenient routes for drug administration. FDA-approved poly (DL-lactide-co-glycolide) (PLGA) acidic nanoparticles (aNPs) have been successfully tested in different models of lysosomal disorders [132–134]. PLGA-aNP traffic to neuronal lysosomes within 24 h from intracerebral injection efficiently restored physiological pH in defective lysosomes, thus ameliorating PD-related dopaminergic neurodegeneration in vitro and in vivo [133]. PLGA-aNPs have also been tested in a *PS1* model, where they rescue lysosomal acidification and autophagy [135]. Lastly, in vivo administration of PLGA-aNPs can prevent α-syn-triggered dopaminergic neurodegeneration in PD mouse model [136]. PLGA-aNP can efficiently counteract lysosomal acidification impairment in fibroblasts of patients affected by X-linked myopathy with excessive autophagy, a genetic disorder related to VMA21, an essential chaperone of the v-ATPase, further strengthening the feasibility of an acidic nanoparticle-based strategy on this class of diseases [133,137].

Another intriguing therapeutical option could arise from recent studies connecting iron homeostasis with lysosomal dysfunction. Iron dyshomeostasis represents an early and crucial step in neurodegeneration, accompanied by accumulation of iron in different neuronal compartments [138,139]. Iron release from the lysosome to the cytoplasm requires acidic pH [140]. Within lysosomes, Fe^{3+} is reduced to Fe^{2+} by the ferric reductase, whose activity depends on acidic pH. Because of its reactivity, cytosolic Fe^{2+} is immediately chelated and excess cytoplasmic iron must be either removed via direct extrusion or neutralized by inclusion in ferritin molecules destined to lysosomal degradation [139]. Impaired lysosomal acidification mediated by both v-ATPase inhibition and genetic depletion of V-ATPase promotes bioavailable iron deficiency resulting in non-apoptotic cell death, mitochondrial dysfunction, and oxidative stress [140–142]. In addition, FAD-associated variants in the amyloid-β protein precursor lead to the inhibition of neuronal v-ATPase, with a subsequent decrease in the bioavailability of Fe^{2+} [143]. Yambire et al. showed that iron deficiency was also able to trigger inflammatory signaling and induce cellular stress response in a Pompe disease brain model and in vitro neuronal models [140]. Remarkably, these pro-inflammatory phenotypes were all corrected by increasing iron concentration in the diet [140], identifying the modulation of iron homeostasis as a potential therapeutic mechanism to contrast v-ATPase dysfunction. Furthermore, treatment with the antioxidant vitamin E (α-tocopherol) significantly reduced iron overload in neurological disorders and counteracted ferroptosis, a type of cell death caused by the accumulation of iron-mediated lipid superoxidation [144–146]. Interestingly, Luthy et al. showed that vitamin E treatment rescued behavioral defects in a knock-in model of *TBC1D24* [85], showing that vitamin E may represent a therapeutical co-adjuvant also in v-ATPase-related brain disorders.

Taken together, these studies emphasize the potential role of selective approaches aiming to counteract lysosomal defects associated with v-ATPase dysfunction (Figure 2 and Table 2).

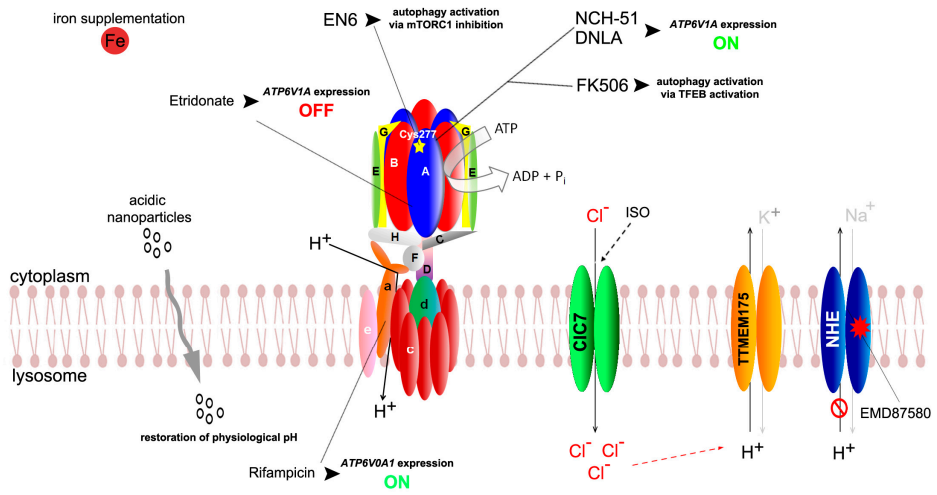


Figure 2. Schematic drawing depicting potential therapeutical approaches for v-ATPase-mediated brain disorders.

Table 2. Potential therapeutical approaches discussed in the study.

Treatment	Target	Mechanism(s) of Action	Tested Models	Ref
EN6	ATP6V1A	Covalent targeting of cysteine 277 of the ATP6V1A subunit and enhancement of autophagy as consequence of mTORC1 inactivation	Mouse and human cells lines; skeletal muscle and heart mouse tissues	[102]
NCH-51	ATP6V1A	Histone deacetylase inhibitor, selective enhancement of *ATP6V1A* expression	Drosophila model and human-induced pluripotent stem cell-derived neurons	[25]

Table 2. *Cont.*

Treatment	Target	Mechanism(s) of Action	Tested Models	Ref
Rifampicin	ATP6V0A1	Enhancement of ATP6V0A1 expression	Immortalized human microglia cells	[106]
FK506 (Tacrolimus)	ATP6V1A	Binding with ATP6V1A followed by TFEB nuclear translocation and activation of autophagy	SH-SYSY neuroblastoma cell line	[108]
Dendrobium nobile Lindl. alkaloids	ATP6V1A	Increased ATP6V1A expression, improved lysosome acidification, and autophagy	amyloid-β and APP/PS1 mice mouse models	[109,110]
Etidronate	ATP6V1A	ATP6V1A inhibition	*Ataxin-2* mouse model	[111]
EMD87580 (rimeporide)	NHE6 channel	Contrasting lysosomal deacidification by preventing proton efflux at the endolysosome	Animal models of myocardial infarction, dystrophic cardiomyopathy, and DMD models; phase I clinical trials in DMD patients	[116–119]
Isoproterenol (ISO)	CIC-7 channel	Contrasting lysosomal deacidification by enhancing parallel Cl^- conductance at the lysosome	*PS1* KO cells; primary fibroblasts derived from *PS1* patients with familial AD	[122]
Poly(DL-lactide-co-glycolide) acidic nanoparticles (PLGA-aNP)	Lysosomes	Direct lowering of lysosomal pH after internalization	Human fibroblasts; BE(2)-M17 neuroblastoma cell line; mouse brain tissues, injected with PD patient-derived Lewy body extracts	[133,135–137]
Iron supplementation	Cellular iron homeostasis	Contrasting bioavailable iron deficiency associated with v-ATPase dysfunction	Yeast lacking v-ATPase components; HEK293T depleted for *ATP6V0C.*; *Gaa*-KO mice	[140–142]

5. Conclusions

Proton-translocating ATPases are involved in transmembrane transport, pH homeostasis, and lysosome acidification. v-ATPases are expressed at high levels in neurons, where they play additional roles, such as neurotransmitter loading into SVs [10,11]. Mutations in different v-ATPase domains and accessory proteins can disrupt proper assembly and function of the pump, leading to impaired cellular processes, such as small molecule transport, protein degradation, intracellular signaling, and autophagy. In the brain, these dysfunctions can also affect development and synaptic function, and contribute to encephalopathy and neurodegeneration in pre-clinical and human models [19,20,25,29,37,38,59,72,147–149] (Figure 3).

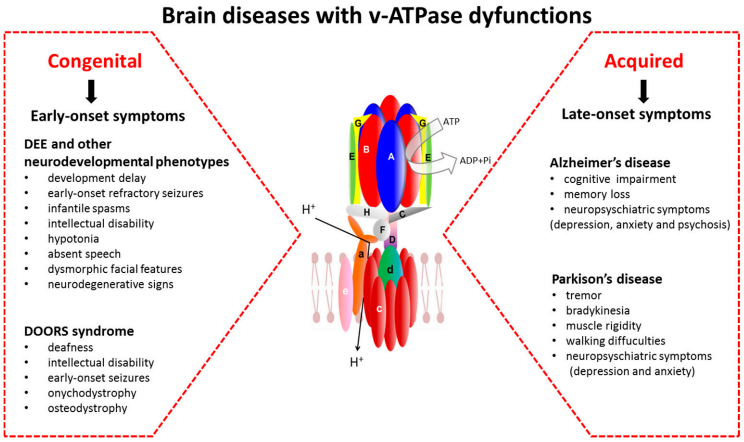


Figure 3. Overview of the clinical manifestations associated with congenital and acquired dysfunction of the v-ATPase in the brain. **Legend**: DEE, developmental and epileptic encephalopathy; DOORS, deafness, onychodystrophy, osteo-dystrophy, retardation and seizures.

This work represents an up-to-date report of the characterized genetic variants of v-ATPase V0 and V1 domains, as well as their accessory proteins ATP6AP2, DMXL2, and TLDc domain-containing proteins. While v-ATPase dysfunction generally leads to pathological conditions, the specific cognitive and neuropathological phenotypes vary among different variants. Further studies are necessary to elucidate the precise mechanisms involved. A deeper understanding of the role of this proton pump in the brain and of its pathogenic variants will aid in the development of precision medicine approaches to improve the clinical symptoms and quality of life in patients with neurological disorders related to v-ATPase dysfunction. Promising potential strategies are now emerging, and future studies will make it possible to define the most appropriate, safe, and effective delivery methods.

Author Contributions: Conceptualization, A.F. (Antonio Falace) and G.V.; methodology, A.F. (Antonio Falace) and G.V.; resources, A.F. (Antonio Falace) and G.V.; data curation, A.F. (Antonio Falace), G.V., M.S., F.Z., A.F. (Anna Fassio) and P.S.; writing—original draft preparation, A.F. (Antonio Falace) and G.V.; visualization, A.F. (Antonio Falace); supervision, A.F. (Anna Fassio), F.Z. and P.S.; project administration, A.F. (Anna Fassio), F.Z. and P.S. All authors have read and agreed to the published version of the manuscript.

Funding: This study was supported by PNRR-MUR-M4C2 PE0000006 (PS), "MNESYS" research program, A multiscale integrate approach to the study of the nervous system in health and disease (PS) and by HUMANITAS MIRASOLE SPA—NET2019.

Acknowledgments: IRCCS 'G. Gaslini' is a member of ERN-Epicare. This work was supported by #NEXTGENERATIONEU (NGEU) and funded by the Ministry of University and Research (MUR), National Recovery and Resilience Plan (NRRP), MNESYS (PE0000006)—A Multiscale Integrated Approach to the Study of the Nervous System in Health and Disease (DN. 1553 11.10.2022).

Conflicts of Interest: P.S. has received speaker fees and participated on advisory boards for Italfarmaco, BioMarin, UCB, Neuraxpharma, and has received research funding from Jazz Pharmaceuticals, Proveca, Italfarmaco. The other authors declare no conflicts of interest.

References

1. Forgac, M. Vacuolar ATPases: Rotary proton pumps in physiology and pathophysiology. *Nat. Rev. Mol. Cell Biol.* **2007**, *8*, 917–929. [CrossRef] [PubMed]
2. Stransky, L.; Cotter, K.; Forgac, M. The Function of V-ATPases in Cancer. *Physiol. Rev.* **2016**, *96*, 1071–1091. [CrossRef]
3. McGuire, C.M.; Forgac, M. Glucose starvation increases V-ATPase assembly and activity in mammalian cells through AMP kinase and phosphatidylinositide 3-kinase/Akt signaling. *J. Biol. Chem.* **2018**, *293*, 9113–9123. [CrossRef]
4. Ratto, E.; Chowdhury, S.R.; Siefert, N.S.; Schneider, M.; Wittmann, M.; Helm, D.; Palm, W. Direct control of lysosomal catabolic activity by mTORC1 through regulation of V-ATPase assembly. *Nat. Commun.* **2022**, *13*, 4848. [CrossRef]
5. Collins, M.P.; Forgac, M. Regulation and function of V-ATPases in physiology and disease. *Biochim. Biophys. Acta (BBA)—Biomembr.* **2020**, *1862*, 183341. [CrossRef]
6. Groszer, M. Rare human ATP6V1A variants provide unique insights into V-ATPase functions. *Brain* **2022**, *145*, 2626–2628. [CrossRef] [PubMed]
7. Nnah, I.C.; Wang, B.; Saqcena, C.; Weber, G.F.; Bonder, E.M.; Bagley, D.; De Cegli, R.; Napolitano, G.; Medina, D.L.; Ballabio, A.; et al. TFEB-driven endocytosis coordinates MTORC1 signaling and autophagy. *Autophagy* **2018**, *15*, 151–164. [CrossRef]
8. Roczniak-Ferguson, A.; Petit, C.S.; Froehlich, F.; Qian, S.; Ky, J.; Angarola, B.; Walther, T.C.; Ferguson, S.M. The Transcription Factor TFEB Links mTORC1 Signaling to Transcriptional Control of Lysosome Homeostasis. *Sci. Signal.* **2012**, *5*, ra42. [CrossRef]
9. Settembre, C.; Zoncu, R.; Medina, D.L.; Vetrini, F.; Erdin, S.; Erdin, S.; Huynh, T.; Ferron, M.; Karsenty, G.; Vellard, M.C.; et al. A lysosome-to-nucleus signalling mechanism senses and regulates the lysosome via mTOR and TFEB. *EMBO J.* **2012**, *31*, 1095–1108. [CrossRef]
10. Puertollano, R.; Ferguson, S.M.; Brugarolas, J.; Ballabio, A. The complex relationship between TFEB transcription factor phosphorylation and subcellular localization. *EMBO J.* **2018**, *37*, e98804. [CrossRef]
11. Napolitano, G.; Ballabio, A. TFEB at a glance. *J. Cell Sci.* **2016**, *129*, 2475–2481. [CrossRef]
12. Sardiello, M.; Palmieri, M.; Di Ronza, A.; Medina, D.L.; Valenza, M.; Gennarino, V.A.; Di Malta, C.; Donaudy, F.; Embrione, V.; Polishchuk, R.S.; et al. A Gene Network Regulating Lysosomal Biogenesis and Function. *Science* **2009**, *325*, 473–477. [CrossRef] [PubMed]
13. Toei, M.; Saum, R.; Forgac, M. Regulation and Isoform Function of the V-ATPases. *Biochemistry* **2010**, *49*, 4715–4723. [CrossRef]

14. Smith, G.A.; Howell, G.J.; Phillips, C.; Muench, S.P.; Ponnambalam, S.; Harrison, M.A. Extracellular and Luminal pH Regulation by Vacuolar H^+-ATPase Isoform Expression and Targeting to the Plasma Membrane and Endosomes. *J. Biol. Chem.* **2016**, *291*, 8500–8515. [CrossRef] [PubMed]
15. Eaton, A.F.; Merkulova, M.; Brown, D. The H^+-ATPase (V-ATPase): From proton pump to signaling complex in health and disease. *Am. J. Physiol. Cell Physiol.* **2021**, *320*, C392–C414. [CrossRef] [PubMed]
16. Gowrisankaran, S.; Milosevic, I. Regulation of synaptic vesicle acidification at the neuronal synapse. *IUBMB Life* **2020**, *72*, 568–576. [CrossRef]
17. Coupland, C.E.; Karimi, R.; Bueler, S.A.; Liang, Y.; Courbon, G.M.; Di Trani, J.M.; Wong, C.J.; Saghian, R.; Youn, J.-Y.; Wang, L.-Y.; et al. High-resolution electron cryomicroscopy of V-ATPase in native synaptic vesicles. *Science* **2024**, *385*, 168–174. [CrossRef]
18. Van Damme, T.; Gardeitchik, T.; Mohamed, M.; Guerrero-Castillo, S.; Freisinger, P.; Guillemyn, B.; Kariminejad, A.; Dalloyaux, D.; van Kraaij, S.; Lefeber, D.J.; et al. Mutations in ATP6V1E1 or ATP6V1A Cause Autosomal-Recessive Cutis Laxa. *Am. J. Hum. Genet.* **2017**, *100*, 216–227. [CrossRef]
19. Fassio, A.; Esposito, A.; Kato, M.; Saitsu, H.; Mei, D.; Marini, C.; Conti, V.; Nakashima, M.; Okamoto, N.; Turker, A.O.; et al. De novo mutations of the ATP6V1A gene cause developmental encephalopathy with epilepsy. *Brain* **2018**, *141*, 1703–1718. [CrossRef]
20. Guerrini, R.; Mei, D.; Kerti-Szigeti, K.; Pepe, S.; Koenig, M.K.; Von Allmen, G.; Cho, M.T.; McDonald, K.; Baker, J.; Bhambhani, V.; et al. Phenotypic and genetic spectrum of ATP6V1A encepha-lopathy: A disorder of lysosomal homeostasis. *Brain* **2022**, *145*, 2687–2703. [CrossRef]
21. Abbas, Y.M.; Wu, D.; Bueler, S.A.; Robinson, C.V.; Rubinstein, J.L. Structure of V-ATPase from the mammalian brain. *Science* **2020**, *367*, 1240–1246. [CrossRef] [PubMed]
22. Wang, H.; Bueler, S.A.; Rubinstein, J.L. Structural basis of V-ATPase V(O) region assembly by Vma12p, 21p, and 22p. *Proc. Natl. Acad. Sci. USA* **2023**, *120*, e2217181120. [CrossRef] [PubMed]
23. Li, B.; Lan, S.; Liu, X.-R.; Ji, J.-J.; He, Y.-Y.; Zhang, D.-M.; Xu, J.; Sun, H.; Shi, Z.; Wang, J.; et al. ATP6V1A variants are associated with childhood epilepsy with favorable outcome. *Seizure* **2024**, *116*, 81–86. [CrossRef]
24. Chen, X.; Lou, Y.; Miao, P.; Cheng, H.; Wan, Z.; Wang, Y.; Yang, F.; Liang, M.; Feng, J. Heterozygous pathogenic variant in the ATP6V1A gene trig-gering epilepsy in a large Chinese pedigree. *Clin. Neurol. Neurosurg.* **2023**, *233*, 107956. [CrossRef]
25. Wang, M.; Li, A.; Sekiya, M.; Beckmann, N.D.; Quan, X.; Schrode, N.; Fernando, M.B.; Yu, A.; Zhu, L.; Cao, J.; et al. Transformative Network Modeling of Multi-omics Data Reveals Detailed Circuits, Key Regulators, and Potential Therapeutics for Alzheimer's Disease. *Neuron* **2020**, *109*, 257–272.e14. [CrossRef]
26. Byrns, C.N.; Bonini, N.M. An Integrated Multi-omics Approach Identifies Therapeutic Potential for ATP6V1A in Late Onset Alzheimer's Disease. *Neuron* **2021**, *109*, 193–194. [CrossRef] [PubMed]
27. Neff, R.A.; Wang, M.; Vatansever, S.; Guo, L.; Ming, C.; Wang, Q.; Wang, E.; Horgusluoglu-Moloch, E.; Song, W.-M.; Li, A.; et al. Molecular subtyping of Alzheimer's disease using RNA sequencing data reveals novel mechanisms and targets. *Sci. Adv.* **2021**, *7*, eabb5398. [CrossRef]
28. Esposito, A.; Pepe, S.; Cerullo, M.S.; Cortese, K.; Semini, H.T.; Giovedì, S.; Guerrini, R.; Benfenati, F.; Falace, A.; Fassio, A. ATP6V1A is required for synaptic rearrangements and plasticity in murine hippocampal neurons. *Acta Physiol.* **2024**, *240*, e14186. [CrossRef]
29. Beauregard-Lacroix, E.; Pacheco-Cuellar, G.; Ajeawung, N.F.; Tardif, J.; Dieterich, K.; Dabir, T.; Vind-Kezunovic, D.; White, S.M.; Zadori, D.; Castiglioni, C.; et al. DOORS syndrome and a recurrent truncating ATP6V1B2 variant. *Genet. Med.* **2021**, *23*, 149–154. [CrossRef]
30. Rousseau, J.; Tadoum, S.B.T.; Jolin, M.L.; Nguyen, T.T.M.; Ajeawung, N.F.; Flenniken, A.M.; Nutter, L.M.J.; Vukobradovic, I.; Rossignol, E.; Campeau, P.M. The *ATP6V1B2* DDOD/DOORS-Associated p.Arg506* Variant Causes Hyperactivity and Seizures in Mice. *Genes* **2023**, *14*, 1538. [CrossRef]
31. Kortüm, F.; Caputo, V.; Bauer, C.K.; Stella, L.; Ciolfi, A.; Alawi, M.; Bocchinfuso, G.; Flex, E.; Paolacci, S.; Dentici, M.L.; et al. Mutations in KCNH1 and ATP6V1B2 cause Zimmermann-Laband syndrome. *Nat. Genet.* **2015**, *47*, 661–667. [CrossRef] [PubMed]
32. Popp, B.; Ekici, A.B.; Thiel, C.T.; Hoyer, J.; Wiesener, A.; Kraus, C.; Reis, A.; Zweier, C. Exome Pool-Seq in neurodevelopmental disorders. *Eur. J. Hum. Genet.* **2017**, *25*, 1364–1376. [CrossRef]
33. Shaw, M.; Winczewska-Wiktor, A.; Badura-Stronka, M.; Koirala, S.; Gardner, A.; Kuszel, Ł.; Kowal, P.; Steinborn, B.; Starczewska, M.; Garry, S.; et al. EXOME REPORT: Novel mutation in ATP6V1B2 segregating with autosomal dominant epilepsy, intellectual disability and mild gingival and nail abnormalities. *Eur. J. Med. Genet.* **2020**, *63*, 103799. [CrossRef] [PubMed]
34. Inuzuka, L.M.; Macedo-Souza, L.I.; Della-Rippa, B.; Monteiro, F.P.; Delgado, D.D.S.; Godoy, L.F.; Ramos, L.; de Athayde Costa, L.S.; Garzon, E.; Kok, F. ATP6V1B2-related epileptic encephalopathy. *Epileptic Disord.* **2020**, *22*, 317–322. [CrossRef] [PubMed]
35. Amore, G.; Calì, E.; Spanò, M.; Ceravolo, G.; Mangano, G.D.; Scorrano, G.; Efthymiou, S.; Salpietro, V.; Houlden, H.; Di Rosa, G. ATP6V1B2-related disorders featuring Lennox-Gastaut-syndrome: A case-based overview. *Brain Dev.* **2023**, *45*, 588–596. [CrossRef] [PubMed]
36. Fernandez-Mosquera, L.; Yambire, K.F.; Couto, R.; Pereyra, L.; Pabis, K.; Ponsford, A.H.; Diogo, C.V.; Stagi, M.; Milosevic, I.; Raimundo, N. Mitochondrial respiratory chain deficiency inhibits lysosomal hydrolysis. *Autophagy* **2019**, *15*, 1572–1591. [CrossRef]

37. Aoto, K.; Kato, M.; Akita, T.; Nakashima, M.; Mutoh, H.; Akasaka, N.; Tohyama, J.; Nomura, Y.; Hoshino, K.; Ago, Y.; et al. *ATP6V0A1* encoding the a1-subunit of the V0 domain of vacuolar H^+-ATPases is essential for brain development in humans and mice. *Nat. Commun.* **2021**, *12*, 2107. [CrossRef]
38. Bott, L.C.; Forouhan, M.; Lieto, M.; Sala, A.J.; Ellerington, R.O.; Johnson, J.A.; Speciale, A.; Criscuolo, C.; Filla, A.; Chitayat, D.; et al. Variants in *ATP6V0A1* cause progressive myoclonus epilepsy and developmental and epileptic encephalopathy. *Brain Commun.* **2021**, *3*, fcab245. [CrossRef]
39. Indrawinata, K.; Argiropoulos, P.; Sugita, S. Structural and functional understanding of disease-associated mutations in V-ATPase subunit a1 and other isoforms. *Front. Mol. Neurosci.* **2023**, *16*, 1135015. [CrossRef]
40. Bagaria, J.; Bagyinszky, E.; An, S.S.A. Genetics, Functions, and Clinical Impact of Presenilin-1 (PSEN1) Gene. *Int. J. Mol. Sci.* **2022**, *23*, 10970. [CrossRef]
41. Lee, J.-H.; Yu, W.H.; Kumar, A.; Lee, S.; Mohan, P.S.; Peterhoff, C.M.; Wolfe, D.M.; Martinez-Vicente, M.; Massey, A.C.; Sovak, G.; et al. Lysosomal Proteolysis and Autophagy Require Presenilin 1 and Are Disrupted by Alzheimer-Related PS1 Mutations. *Cell* **2010**, *141*, 1146–1158. [CrossRef] [PubMed]
42. Im, E.; Jiang, Y.; Stavrides, P.H.; Darji, S.; Erdjument-Bromage, H.; Neubert, T.A.; Choi, J.Y.; Wegiel, J.; Lee, J.H.; Nixon, R.A. Lysosomal dysfunction in Down syndrome and Alzheimer mouse models is caused by v-ATPase inhibition by Tyr682-phosphorylated APP βCTF. *Sci. Adv.* **2023**, *9*, eadg1925. [CrossRef] [PubMed]
43. Lott, I.T.; Head, E. Dementia in Down syndrome: Unique insights for Alzheimer disease research. *Nat. Rev. Neurol.* **2019**, *15*, 135–147. [CrossRef] [PubMed]
44. Wallings, R.; Connor-Robson, N.; Wade-Martins, R. LRRK2 interacts with the vacuolar-type H^+-ATPase pump a1 subunit to regulate lysosomal function. *Hum. Mol. Genet.* **2019**, *28*, 2696–2710. [CrossRef]
45. Bagh, M.B.; Peng, S.; Chandra, G.; Zhang, Z.; Singh, S.P.; Pattabiraman, N.; Liu, A.; Mukherjee, A.B. Misrouting of v-ATPase subunit V0a1 dysregulates lysosomal acidification in a neurodegenerative lysosomal storage disease model. *Nat. Commun.* **2017**, *8*, 14612. [CrossRef]
46. Wang, L.; Wu, D.; Robinson, C.V.; Wu, H.; Fu, T.M. Structures of a complete human V-ATPase reveal mechanisms of its assembly. *Mol. Cell* **2020**, *80*, 501–511. [CrossRef]
47. Mucha, B.E.; Banka, S.; Ajeawung, N.F.; Molidperee, S.; Chen, G.G.; Koenig, M.K.; Adejumo, R.B.; Till, M.; Harbord, M.; Perrier, R.; et al. A new microdeletion syndrome involving TBC1D24, ATP6V0C, and PDPK1 causes epilepsy, microcephaly, and developmental delay. *Genet. Med.* **2019**, *21*, 1058–1064. [CrossRef]
48. Tinker, R.J.; Burghel, G.J.; Garg, S.; Steggall, M.; Cuvertino, S.; Banka, S. Haploinsufficiency of ATP6V0C possibly underlies 16p13.3 deletions that cause microcephaly, seizures, and neurodevelopmental disorder. *Am. J. Med. Genet. A* **2020**, *185*, 196–202. [CrossRef]
49. Ittiwut, C.; Poonmaksatit, S.; Boonsimma, P.; Desudchit, T.; Suphapeetiporn, K.; Ittiwut, R.; Shotelersuk, V. Novel *de novo* mutation substantiates ATP6V0C as a gene causing epilepsy with intellectual disability. *Brain Dev.* **2020**, *43*, 490–494. [CrossRef]
50. Tian, Y.; Zhai, Q.-X.; Li, X.-J.; Shi, Z.; Cheng, C.-F.; Fan, C.-X.; Tang, B.; Zhang, Y.; He, Y.-Y.; Li, W.-B.; et al. ATP6V0C Is Associated with Febrile Seizures and Epilepsy with Febrile Seizures Plus. *Front. Mol. Neurosci.* **2022**, *15*, 889534. [CrossRef]
51. Zhao, S.; Zhang, X.; Yang, L.; Wang, Y.; Jia, S.; Li, X.; Wang, Z.; Yang, F.; Liang, M.; Wang, X.; et al. ATP6V0C gene variants were identified in individuals with epilepsy, with or without developmental delay. *J. Hum. Genet.* **2023**, *68*, 589–597. [CrossRef] [PubMed]
52. Mattison, K.; Tossing, G.; Mulroe, F.; Simmons, C.; Butler, K.M.; Schreiber, A.; Alsadah, A.E.; Neilson, D.; Naess, K.; Wedell, A.; et al. *ATP6V0C* variants impair V-ATPase function causing a neurodevelopmental disorder often associated with epilepsy. *Brain* **2022**, *146*, 1357–1372. [CrossRef] [PubMed]
53. Hoffmann, N.; Peters, J. Functions of the (pro)renin receptor (Atp6ap2) at molecular and system levels: Pathological implications in hypertension, renal and brain development, inflammation, and fibrosis. *Pharmacol. Res.* **2021**, *173*, 105922. [CrossRef]
54. Guida, M.C.; Hermle, T.; Graham, L.A.; Hauser, V.; Ryan, M.; Stevens, T.H.; Simons, M.; Ma, M.; Kumar, S.; Purushothaman, L.; et al. ATP6AP2 functions as a V-ATPase assembly factor in the endoplasmic reticulum. *Mol. Biol. Cell* **2018**, *29*, 2156–2164. [CrossRef]
55. Halbach, O.v.B.U.; Bracke, A. Roles and functions of Atp6ap2 in the brain. *Neural Regen. Res.* **2018**, *13*, 2038–2043. [CrossRef] [PubMed]
56. Dulac, A.; Issa, A.-R.; Sun, J.; Matassi, G.; Jonas, C.; Rahmani, Z.; Chérif-Zahar, B.; Cattaert, D.; Birman, S. A Novel Neuron-Specific Regulator of the V-ATPase in Drosophila. *eNeuro* **2021**, *8*, 1–22. [CrossRef]
57. Hedera, P.; Alvarado, D.; Beydoun, A.; Fink, J.K. Novel mental retardation-epilepsy syndrome linked to Xp21.1-p11. *Ann. Neurol.* **2002**, *51*, 45–50. [CrossRef]
58. Ramser, J.; Abidi, F.E.; Burckle, C.A.; Lenski, C.; Toriello, H.; Wen, G.; Lubs, H.A.; Engert, S.; Stevenson, R.E.; Meindl, A.; et al. A unique exonic splice enhancer mutation in a family with X-linked mental retardation and epilepsy points to a novel role of the renin receptor. *Hum. Mol. Genet.* **2005**, *14*, 1019–1027. [CrossRef]
59. Hirose, T.; Cabrera-Socorro, A.; Chitayat, D.; Lemonnier, T.; Féraud, O.; Cifuentes-Diaz, C.; Gervasi, N.; Mombereau, C.; Ghosh, T.; Stoica, L.; et al. ATP6AP2 variant impairs CNS development and neuronal survival to cause fulminant neurodegeneration. *J. Clin. Investig.* **2019**, *129*, 2145–2162. [CrossRef]

60. Dubos, A.; Castells-Nobau, A.; Meziane, H.; Oortveld, M.A.; Houbaert, X.; Iacono, G.; Martin, C.; Mittelhaeuser, C.; Lalanne, V.; Kramer, J.M.; et al. Conditional depletion of intellectual disability and Parkinsonism candidate gene ATP6AP2 in fly and mouse induces cognitive impairment and neurodegeneration. *Hum. Mol. Genet.* **2015**, *24*, 6736–6755. [CrossRef]
61. Poorkaj, P.; Raskind, W.H.; Leverenz, J.B.; Matsushita, M.; Zabetian, C.P.; Samii, A.; Kim, S.; Gazi, N.; Nutt, J.G.; Wolff, J.; et al. A novel X-linked four-repeat tauopathy with Parkinsonism and spasticity. *Mov. Disord.* **2010**, *25*, 1409–1417. [CrossRef] [PubMed]
62. Korvatska, O.; Strand, N.S.; Berndt, J.D.; Strovas, T.; Chen, D.-H.; Leverenz, J.B.; Kiianitsa, K.; Mata, I.F.; Karakoc, E.; Greenup, J.L.; et al. Altered splicing of ATP6AP2 causes X-linked parkinsonism with spasticity (XPDS). *Hum. Mol. Genet.* **2013**, *22*, 3259–3268. [CrossRef] [PubMed]
63. Edelman, W.C.; Kiianitsa, K.; Virmani, T.; Martinez, R.A.; Young, J.E.; Keene, C.D.; Bird, T.D.; Raskind, W.H.; Korvatska, O. Reduced gene dosage is a common mechanism of neuropathologies caused by ATP6AP2 splicing mutations. *Park. Relat. Disord.* **2022**, *101*, 31–38. [CrossRef] [PubMed]
64. Nagano, F.; Kawabe, H.; Nakanishi, H.; Shinohara, M.; Deguchi-Tawarada, M.; Takeuchi, M.; Sasaki, T.; Takai, Y. Rabconnectin-3, a Novel Protein That Binds Both GDP/GTP Exchange Protein and GTPase-activating Protein for Rab3 Small G Protein Family. *J. Biol. Chem.* **2002**, *277*, 9629–9632. [CrossRef]
65. Kawabe, H.; Sakisaka, T.; Yasumi, M.; Shingai, T.; Izumi, G.; Nagano, F.; Deguchi-Tawarada, M.; Takeuchi, M.; Nakanishi, H.; Takai, Y. A novel rabconnectin-3-binding protein that directly binds a GDP/GTP exchange protein for Rab3A small G protein implicated in Ca(2+)-dependent exocytosis of neuro-transmitter. *Genes Cells* **2003**, *8*, 537–546. [CrossRef]
66. Tata, B.; Huijbregts, L.; Jacquier, S.; Csaba, Z.; Genin, E.; Meyer, V.; Leka, S.; Dupont, J.; Charles, P.; Chevenne, D.; et al. Haploinsufficiency of Dmxl2, Encoding a Synaptic Protein, Causes Infertility Associated with a Loss of GnRH Neurons in Mouse. *PLoS Biol.* **2014**, *12*, e1001952. [CrossRef]
67. Gobé, C.; Elzaiat, M.; Meunier, N.; André, M.; Sellem, E.; Congar, P.; Jouneau, L.; Allais-Bonnet, A.; Naciri, I.; Passet, B.; et al. Dual role of DMXL2 in olfactory information transmission and the first wave of spermatogenesis. *PLoS Genet.* **2019**, *15*, e1007909. [CrossRef]
68. Yan, Y.; Denef, N.; Schüpbach, T. The Vacuolar Proton Pump, V-ATPase, Is Required for Notch Signaling and Endosomal Trafficking in Drosophila. *Dev. Cell* **2009**, *17*, 387–402. [CrossRef] [PubMed]
69. Einhorn, Z.; Trapani, J.G.; Liu, Q.; Nicolson, T. Rabconnectin3 promotes stable activity of the HC pump on synaptic vesicles in hair cells. *J. Neurosci.* **2012**, *32*, 11144–11156. [CrossRef]
70. Tuttle, A.M.; Hoffman, T.L.; Schilling, T.F. Rabconnectin-3a Regulates Vesicle Endocytosis and Canonical Wnt Signaling in Zebrafish Neural Crest Migration. *PLoS Biol.* **2014**, *12*, e1001852. [CrossRef]
71. Merkulova, M.; Păunescu, T.G.; Azroyan, A.; Marshansky, V.; Breton, S.; Brown, D. Mapping the $H^{(+)}$ (V)-ATPase interactome: Identification of proteins involved in trafficking, folding, assembly and phosphorylation. *Sci. Rep.* **2015**, *5*, 14827. [CrossRef] [PubMed]
72. Esposito, A.; Falace, A.; Wagner, M.; Gal, M.; Mei, D.; Conti, V.; Pisano, T.; Aprile, D.; Cerullo, M.S.; De Fusco, A.; et al. Biallelic DMXL2 mutations impair autophagy and cause Ohtahara syndrome with progressive course. *Brain* **2019**, *142*, 3876–3891. [CrossRef] [PubMed]
73. Maddirevula, S.; Alzahrani, F.; Al-Owain, M.; Al Muhaizea, M.A.; Kayyali, H.R.; AlHashem, A.; Rahbeeni, Z.; Al-Otaibi, M.; Alzaidan, H.I.; Balobaid, A.; et al. Autozygome and high throughput confirmation of disease genes candidacy. *Genet. Med.* **2018**, *21*, 736–742. [CrossRef]
74. Kannan, M.; Bayam, E.; Wagner, C.; Rinaldi, B.; Kretz, P.F.; Tilly, P.; Roos, M.; McGillewie, L.; Bär, S.; Minocha, S.; et al. WD40-repeat 47, a microtubule-associated protein, is essential for brain development and autophagy. *Proc. Natl. Acad. Sci. USA* **2017**, *114*, E9308–E9317. [CrossRef]
75. Wilkens, S.; Khan, M.; Knight, K.; Oot, R.A. Tender love and disassembly: How a TLDc domain protein breaks the VATPase. *BioEssays* **2023**, *45*, e2200251. [CrossRef] [PubMed]
76. Castroflorio, E.; Hoed, J.D.; Svistunova, D.; Finelli, M.J.; Cebrian-Serrano, A.; Corrochano, S.; Bassett, A.R.; Davies, B.; Oliver, P.L. The Ncoa7 locus regulates V-ATPase formation and function, neurodevelopment and behaviour. *Cell. Mol. Life Sci.* **2020**, *78*, 3503–3524. [CrossRef]
77. Eaton, A.F.; Brown, D.; Merkulova, M. The evolutionary conserved TLDc domain defines a new class of (H^+)V-ATPase interacting proteins. *Sci. Rep.* **2021**, *11*, 22654. [CrossRef]
78. Oot, R.A.; Wilkens, S. Human V-ATPase function is positively and negatively regulated by TLDc proteins. *Structure* **2024**, *32*, 989–1000.e6. [CrossRef]
79. Corbett, M.A.; Bahlo, M.; Jolly, L.; Afawi, Z.; Gardner, A.E.; Oliver, K.L.; Tan, S.; Coffey, A.; Mulley, J.C.; Dibbens, L.M.; et al. A Focal Epilepsy and Intellectual Disability Syndrome Is Due to a Mutation in TBC1D. *Am. J. Hum. Genet.* **2010**, *87*, 371–375. [CrossRef]
80. Falace, A.; Filipello, F.; La Padula, V.; Vanni, N.; Madia, F.; Tonelli, D.D.P.; De Falco, F.A.; Striano, P.; Bricarelli, F.D.; Minetti, C.; et al. TBC1D24, an ARF6-Interacting Protein, Is Mutated in Familial Infantile Myoclonic Epilepsy. *Am. J. Hum. Genet.* **2010**, *87*, 365–370. [CrossRef]
81. Afawi, Z.; Mandelstam, S.; Korczyn, A.D.; Kivity, S.; Walid, S.; Shalata, A.; Oliver, K.L.; Corbett, M.; Gecz, J.; Berkovic, S.F.; et al. TBC1D24 mutation associated with focal epilepsy, cog-nitive impairment and a distinctive cerebro-cerebellar malformation. *Epilepsy Res.* **2013**, *105*, 240–244. [CrossRef] [PubMed]

82. Guven, A.; Tolun, A. *TBC1D24* truncating mutation resulting in severe neurodegeneration. *J. Med. Genet.* **2013**, *50*, 199–202. [CrossRef] [PubMed]
83. Milh, M.; Falace, A.; Villeneuve, N.; Vanni, N.; Cacciagli, P.; Assereto, S.; Nabbout, R.; Benfenati, F.; Zara, F.; Chabrol, B.; et al. Novel compound heterozygous mutations in TBC1D24 cause familial malignant migrating partial seizures of infancy. *Hum. Mutat.* **2013**, *34*, 869–872. [CrossRef]
84. Balestrini, S.; Milh, M.; Castiglioni, C.; Lüthy, K.; Finelli, M.J.; Verstreken, P.; Cardon, A.; Stražišar, B.G.; Holder, J.L., Jr.; Lesca, G.; et al. TBC1D24 genotype-phenotype correlation: Epilepsies and other neurologic features. *Neurology* **2016**, *87*, 77–85. [CrossRef]
85. Lüthy, K.; Mei, D.; Fischer, B.; De Fusco, M.; Swerts, J.; Paesmans, J.; Parrini, E.; Lubarr, N.A.; Meijer, I.; Mackenzie, K.M.; et al. TBC1D24-TLDc-related epilepsy exercise-induced dystonia: Rescue by antioxidants in a disease model. *Brain* **2019**, *142*, 2319–2335. [CrossRef]
86. Nakashima, M.; Negishi, Y.; Hori, I.; Hattori, A.; Saitoh, S.; Saitsu, H. A case of early-onset epileptic encephalopathy with a ho-mozygous TBC1D24 variant caused by uniparental isodisomy. *Am. J. Med. Genet. A* **2019**, *179*, 645–649. [CrossRef]
87. Campeau, P.M.; Kasperaviciute, D.; Lu, J.T.; Burrage, L.C.; Kim, C.; Hori, M.; Powell, B.R.; Stewart, F.; Félix, T.M.; van den Ende, J.; et al. The genetic basis of DOORS syndrome: An exome-sequencing study. *Lancet Neurol.* **2014**, *13*, 44–58. [CrossRef]
88. Gao, X.; Dai, P.; Yuan, Y.-Y. Genetic architecture and phenotypic landscape of deafness and onychodystrophy syndromes. *Hum. Genet.* **2021**, *141*, 821–838. [CrossRef] [PubMed]
89. Falace, A.; Buhler, E.; Fadda, M.; Watrin, F.; Lippiello, P.; Pallesi-Pocachard, E.; Baldelli, P.; Benfenati, F.; Zara, F.; Represa, A.; et al. TBC1D24 regulates neuronal migration and maturation through modulation of the ARF6-dependent pathway. *Proc. Natl. Acad. Sci. USA* **2014**, *111*, 2337–2342. [CrossRef]
90. Aprile, D.; Fruscione, F.; Baldassari, S.; Fadda, M.; Ferrante, D.; Falace, A.; Buhler, E.; Sartorelli, J.; Represa, A.; Baldelli, P.; et al. TBC1D24 regulates axonal outgrowth and membrane trafficking at the growth cone in rodent and human neurons. *Cell Death Differ.* **2019**, *26*, 2464–2478. [CrossRef]
91. Finelli, M.J.; Sanchez-Pulido, L.; Liu, K.X.; Davies, K.E.; Oliver, P.L. The Evolutionarily Conserved Tre2/Bub2/Cdc16 (TBC), Lysin Motif (LysM), Domain Catalytic (TLDc) Domain Is Neuroprotective against Oxidative Stress. *J. Biol. Chem.* **2016**, *291*, 2751–2763. [CrossRef]
92. Fernandes, A.C.; Uytterhoeven, V.; Kuenen, S.; Wang, Y.C.; Slabbaert, J.R.; Swerts, J.; Kasprowicz, J.; Aerts, S.; Verstreken, P. Reduced synaptic vesicle protein degradation at lysosomes curbs TBC1D24/sky-induced neurodegeneration. *J. Cell Biol.* **2014**, *207*, 453–462. [CrossRef]
93. Lin, L.; Lyu, Q.; Kwan, P.-Y.; Zhao, J.; Fan, R.; Chai, A.; Lai, C.S.W.; Chan, Y.-S.; Shen, X.; Lai, K.-O. The epilepsy and intellectual disability-associated protein TBC1D24 regulates the maintenance of excitatory synapses and animal behaviors. *PLoS Genet.* **2020**, *16*, e1008587. [CrossRef]
94. Doan, R.N.; Lim, E.T.; De Rubeis, S.; Betancur, C.; Cutler, D.J.; Chiocchetti, A.G.; Overman, L.M.; Soucy, A.; Goetze, S. Recessive gene disruptions in autism spectrum disorder. *Nat. Genet.* **2019**, *51*, 1092–1098. [CrossRef] [PubMed]
95. Liu, K.X.; Edwards, B.; Lee, S.; Finelli, M.J.; Davies, B.; Davies, K.E.; Oliver, P.L. Neuron-specific antioxidant OXR1 extends survival of a mouse model of amyotrophic lateral sclerosis. *Brain* **2015**, *138*, 1167–1181. [CrossRef] [PubMed]
96. Finelli, M.J.; Oliver, P.L. TLDc proteins: New players in the oxidative stress response and neurological disease. *Mamm. Genome* **2017**, *28*, 395–406. [CrossRef]
97. Bucknor, E.M.; Johnson, E.; Efthymiou, S.; Alvi, J.R.; Sultan, T.; Houlden, H.; Maroofian, R.; Karimiani, E.G.; Finelli, M.J.; Oliver, P.L. Neuroinflammation and Lysosomal Abnormalities Characterise the Essential Role for Oxidation Resistance 1 in the Developing and Adult Cerebellum. *Antioxidants* **2024**, *13*, 685. [CrossRef]
98. Khan, M.; Lee, S.; Couoh-Cardel, S.A.; Oot, R.; Kim, H.; Wilkens, S.; Roh, S. Oxidative stress protein Oxr1 promotes V-ATPase holoenzyme disassembly in catalytic activity-independent manner. *EMBO J.* **2021**, *41*, e109360. [CrossRef]
99. Wang, J.; Rousseau, J.; Kim, E.; Ehresmann, S.; Cheng, Y.-T.; Duraine, L.; Zuo, Z.; Park, Y.-J.; Li-Kroeger, D.; Bi, W.; et al. Loss of Oxidation Resistance 1, OXR1, Is Associated with an Autosomal-Recessive Neurological Disease with Cerebellar Atrophy and Lysosomal Dysfunction. *Am. J. Hum. Genet.* **2019**, *105*, 1237–1253. [CrossRef]
100. Lin, X.; Wang, W.; Yang, M.; Damseh, N.; de Sousa, M.M.L.; Jacob, F.; Lång, A.; Kristiansen, E.; Pannone, M.; Kissova, M.; et al. A loss-of-function mutation in human Oxidation Resistance 1 disrupts the spatial–temporal regulation of histone arginine methylation in neurodevelopment. *Genome Biol.* **2023**, *24*, 216. [CrossRef]
101. Milgrom, E.; Diab, H.; Middleton, F.; Kane, P.M. Loss of Vacuolar Proton-translocating ATPase Activity in Yeast Results in Chronic Oxidative Stress. *J. Biol. Chem.* **2007**, *282*, 7125–7136. [CrossRef]
102. Chung, C.Y.-S.; Shin, H.R.; Berdan, C.A.; Ford, B.; Ward, C.C.; Olzmann, J.A.; Zoncu, R.; Nomura, D.K. Covalent targeting of the vacuolar H^+-ATPase activates autophagy via mTORC1 inhibition. *Nat. Chem. Biol.* **2019**, *15*, 776–785. [CrossRef] [PubMed]
103. Kilic, U.; Kilic, E.; Lingor, P.; Yulug, B.; Bahr, M. Rifampicin inhibits neurodegeneration in the optic nerve transection model in vivo and after 1-methyl-4-phenylpyridinium intoxication in vitro. *Acta Neuropathol.* **2004**, *108*, 65–68. [CrossRef] [PubMed]
104. Jing, X.; Shi, Q.; Bi, W.; Zeng, Z.; Liang, Y.; Wu, X.; Xiao, S.; Liu, J.; Yang, L.; Tao, E. Rifampicin protects PC12 cells from rotenone-induced cytotoxicity by activating GRP78 via PERK-eIF2alpha-ATF4 pathway. *PLoS ONE* **2014**, *9*, e92110. [CrossRef]

105. Wu, X.; Liang, Y.; Jing, X.; Lin, D.; Chen, Y.; Zhou, T.; Peng, S.; Zheng, D.; Zeng, Z.; Lei, M.; et al. Rifampicin Prevents SH-SY5Y Cells from Rotenone-Induced Apoptosis via the PI3K/Akt/GSK-3β/CREB Signaling Pathway. *Neurochem. Res.* **2018**, *43*, 886–893. [CrossRef] [PubMed]
106. Liang, Y.; Zheng, D.; Peng, S.; Lin, D.; Jing, X.; Zeng, Z.; Chen, Y.; Huang, K.; Xie, Y.; Zhou, T.; et al. Rifampicin attenuates rotenone-treated microglia inflammation via improving lysosomal function. *Toxicol. Vitr.* **2020**, *63*, 104690. [CrossRef]
107. Dumont, F.J. FK506, An Immunosuppressant Targeting Calcineurin Function. *Curr. Med. Chem.* **2000**, *7*, 731–748. [CrossRef]
108. Kim, D.; Hwang, H.-Y.; Kim, J.Y.; Lee, J.Y.; Yoo, J.S.; Marko-Varga, G.; Kwon, H.J. FK506, an Immunosuppressive Drug, Induces Autophagy by Binding to the V-ATPase Catalytic Subunit A in Neuronal Cells. *J. Proteome Res.* **2016**, *16*, 55–64. [CrossRef]
109. Li, L.S.; Lu, Y.L.; Nie, J.; Xu, Y.Y.; Zhang, W.; Yang, W.J.; Gong, Q.H.; Lu, Y.F.; Lu, Y.; Shi, J.S. Dendrobium nobile Lindl alkaloid, a novel autophagy inducer, protects against axonal degeneration induced by Aβ25–35 in hippocampus neurons in vitro. *CNS Neurosci. Ther.* **2017**, *23*, 329. [CrossRef]
110. Nie, J.; Jiang, L.S.; Zhang, Y.; Tian, Y.; Li, L.S.; Lu, Y.L.; Yang, W.J.; Shi, J.S. Dendrobium nobile Lindl. Alkaloids Decreases the Level of Intracellular β-Amyloid by Improving Impaired Autolysosomal Proteolysis in APP/PS1 Mice. *Front. Pharmacol.* **2018**, *18*, 1479. [CrossRef]
111. Kim, G.; Nakayama, L.; Blum, J.A.; Akiyama, T.; Boeynaems, S.; Chakraborty, M.; Couthouis, J.; Tassoni-Tsuchida, E.; Rodriguez, C.M.; Bassik, M.C.; et al. Genome-wide CRISPR screen reveals v-ATPase as a drug target to lower levels of ALS protein ataxin-2. *Cell Rep.* **2022**, *41*, 111508. [CrossRef] [PubMed]
112. Drake, M.T.; Clarke, B.L.; Khosla, S. Bisphosphonates: Mechanism of Action and Role in Clinical Practice. *Mayo Clin. Proc.* **2008**, *83*, 1032–1045. [CrossRef] [PubMed]
113. Zhou, Z.; Bai, J.; Zhong, S.; Zhang, R.; Kang, K.; Zhang, X.; Xu, Y.; Zhao, C.; Zhao, M. Downregulation of ATP6V1A Involved in Alzheimer's Disease via Synaptic Vesicle Cycle, Phagosome, and Oxidative Phosphorylation. *Oxid. Med. Cell. Longev.* **2021**, *2021*, 5555634. [CrossRef] [PubMed]
114. Zhao, H.; Carney, K.E.; Falgoust, L.; Pan, J.W.; Sun, D.; Zhang, Z. Emerging roles of Na^+/H^+ exchangers in epilepsy and developmental brain disorders. *Prog. Neurobiol.* **2016**, *138–140*, 19–35. [CrossRef]
115. Kondapalli, K.C.; Prasad, H.; Rao, R. An inside job: How endosomal Na^+/H^+ exchangers link to autism and neurological disease. *Front. Cell. Neurosci.* **2014**, *23*, 172. [CrossRef]
116. Previtali, S.C.; Gidaro, T.; Díaz-Manera, J.; Zambon, A.; Carnesecchi, S.; Roux-Lombard, P.; Spitali, P.; Signorelli, M.; Szigyarto, C.A.-K.; Johansson, C.; et al. Rimeporide as a first- in-class NHE-1 inhibitor: Results of a phase Ib trial in young patients with Duchenne Muscular Dystrophy. *Pharmacol. Res.* **2020**, *159*, 104999. [CrossRef]
117. Englhardt, S.; Hein, L.; Keller, U.; Klämbt, K.; Lohse, M.J. Inhibition of Na^+-H^+ exchange prevents hypertrophy, fibrosis, and heart failure in beta(1)-adrenergic receptor transgenic mice. *Circ. Res.* **2002**, *90*, 814–819. [CrossRef]
118. Xian, X.; Pohlkamp, T.; Durakoglugil, M.S.; Wong, C.H.; Beck, J.K.; Lane-Donovan, C.; Plattner, F.; Herz, J. Reversal of ApoE4-induced recycling block as a novel prevention approach for Alzheimer's disease. *eLife* **2018**, *7*, 40048. [CrossRef]
119. Metwally, S.A.H.; Paruchuri, S.S.; Yu, L.; Capuk, O.; Pennock, N.; Sun, D.; Song, S. Pharmacological Inhibition of NHE1 Protein Increases White Matter Resilience and Neurofunctional Recovery after Ischemic Stroke. *Int. J. Mol. Sci.* **2023**, *24*, 13289. [CrossRef]
120. Weinert, S.; Jabs, S.; Supanchart, C.; Schweizer, M.; Gimber, N.; Richter, M.; Rademann, J.; Stauber, T.; Kornak, U.; Jentsch, T.J. Lysosomal pathology and osteopetrosis upon loss of H^+-driven lysosomal Cl^- accumulation. *Science* **2010**, *328*, 1401–1403. [CrossRef]
121. Zifarelli, G. The Role of the Lysosomal Cl^-/H^+ Antiporter ClC-7 in Osteopetrosis and Neurodegeneration. *Cells* **2022**, *11*, 366. [CrossRef] [PubMed]
122. Lee, J.-H.; Wolfe, D.M.; Darji, S.; McBrayer, M.K.; Colacurcio, D.J.; Kumar, A.; Stavrides, P.; Mohan, P.S.; Nixon, R.A. β2-adrenergic Agonists Rescue Lysosome Acidification and Function in PSEN1 Deficiency by Reversing Defective ER-to-lysosome Delivery of ClC-7. *Mol. Biol.* **2020**, *432*, 2633–2650. [CrossRef] [PubMed]
123. O'Shaughnessy, K.M. Chapter 23—Adrenergic mechanisms and drugs. In *Clinical Pharmacology*, 11th ed.; Bennett, P.N., Brown, M.J., Sharma, P., Eds.; Churchill Livingstone: London, UK, 2012; pp. 382–392.
124. Wu, L.; Lin, Y.; Song, J.; Li, L.; Rao, X.; Wan, W.; Wei, G.; Hua, F.; Ying, J. TMEM175: A lysosomal ion channel associated with neurological diseases. *Neurobiol. Dis.* **2023**, *185*, 106244. [CrossRef]
125. Jinn, S.; Blauwendraat, C.; Toolan, D.A.; Gretzula, C.E.; Drolet, R.; Smith, S.A.; Nalls, M.; Marcus, J.; Singleton, A.B.; Stone, D.J. Functionalization of the TMEM175 p.M393T variant as a risk factor for Parkinson disease. *Hum. Mol. Genet.* **2019**, *28*, 3244–3254. [CrossRef]
126. Rudakou, U.; Yu, E.; Krohn, L.; Ruskey, J.A.; Asayesh, F.; Dauvilliers, Y.; Spiegelman, D.; Greenbaum, L.; Fahn, S.; Waters, C.H.; et al. Targeted sequencing of Parkinson's disease loci genes highlights SYT11, FGF20 and other associations. *Brain* **2021**, *144*, 462–472. [CrossRef]
127. Wie, J.; Liu, Z.; Song, H.; Tropea, T.F.; Yang, L.; Wang, H.; Liang, Y.; Cang, C.; Aranda, K.; Lohmann, J.; et al. A growth-factor-activated lysosomal K^+ channel regulates Parkinson's pathology. *Nature* **2021**, *591*, 431–437. [CrossRef] [PubMed]
128. Grover, S.; Kumar Sreelatha, A.A.; Pihlstrom, L.; Domenighetti, C.; Schulte, C.; Sugier, P.E.; Radivojkov-Blagojevic, M.; Lichtner, P.; Mohamed, O.; Portugal, B.; et al. Genome-wide Association and Meta-analysis of Age at Onset in Parkinson Disease: Evidence from the COURAGE-PD Consortium. *Neurology* **2022**, *99*, e698–e710. [CrossRef]

129. Oh, S.; Stix, R.; Zhou, W.; Faraldo-Gómez, J.D.; Hite, R.K. Mechanism of 4-aminopyridine inhibition of the lysosomal channel TMEM. *Proc. Natl. Acad. Sci. USA* **2022**, *119*, e2208882119. [CrossRef]
130. Brunner, J.D.; Jakob, R.P.; Schulze, T.; Neldner, Y.; Moroni, A.; Thiel, G.; Maier, T.; Schenck, S. Structural basis for ion selectivity in TMEM175 K^+ channels. *eLife* **2020**, *9*, 53683. [CrossRef]
131. Guo, Z.-H.; Khattak, S.; Rauf, M.A.; Ansari, M.A.; Alomary, M.N.; Razak, S.; Yang, C.-Y.; Wu, D.-D.; Ji, X.-Y. Role of Nanomedicine-Based Therapeutics in the Treatment of CNS Disorders. *Molecules* **2023**, *28*, 1283. [CrossRef]
132. Sahay, G.; Alakhova, D.Y.; Kabanov, A.V. Endocytosis of nanomedicines. *J. Control. Release* **2010**, *145*, 182–195. [CrossRef] [PubMed]
133. Bourdenx, M.; Daniel, J.; Genin, E.; Soria, F.N.; Blanchard-Desce, M.; Bezard, E.; Dehay, B. Nanoparticles restore lysosomal acidification defects: Implications for Parkinson and other lysosomal-related diseases. *Autophagy* **2016**, *12*, 472–483. [CrossRef] [PubMed]
134. Cunha, A.; Gaubert, A.; Latxague, L.; Dehay, B. PLGA-Based Nanoparticles for Neuroprotective Drug Delivery in Neurodegenerative Diseases. *Pharmaceutics* **2021**, *13*, 1042. [CrossRef] [PubMed]
135. Lee, J.-H.; McBrayer, M.K.; Wolfe, D.M.; Haslett, L.J.; Kumar, A.; Sato, Y.; Lie, P.P.; Mohan, P.; Coffey, E.E.; Kompella, U.; et al. Presenilin 1 Maintains Lysosomal Ca^{2+} Homeostasis via TRPML1 by Regulating vATPase-Mediated Lysosome Acidification. *Cell Rep.* **2015**, *12*, 1430–1444. [CrossRef]
136. Arotcarena, M.; Soria, F.N.; Cunha, A.; Doudnikoff, E.; Prévot, G.; Daniel, J.; Blanchard-Desce, M.; Barthélémy, P.; Bezard, E.; Crauste-Manciet, S.; et al. Acidic nanoparticles protect against α-synuclein-induced neurodegeneration through the restoration of lysosomal function. *Aging Cell* **2022**, *21*, e13584. [CrossRef]
137. Ramachandran, N.; Munteanu, I.; Wang, P.; Ruggieri, A.; Rilstone, J.J.; Israelian, N.; Naranian, T.; Paroutis, P.; Guo, R.; Ren, Z.-P.; et al. VMA21 deficiency prevents vacuolar ATPase assembly and causes autophagic vacuolar myopathy. *Acta Neuropathol.* **2013**, *125*, 439–457. [CrossRef]
138. Ndayisaba, A.; Kaindlstorfer, C.; Wenning, G.K. Iron in Neurodegeneration—Cause or Consequence? *Front. Neurosci.* **2019**, *13*, 180. [CrossRef]
139. Rizzollo, F.; More, S.; Vangheluwe, P.; Agostinis, P. The lysosome as a master regulator of iron metabolism. *Trends Biochem. Sci.* **2021**, *46*, 960–975. [CrossRef]
140. Yambire, K.F.; Rostosky, C.; Watanabe, T.; Pacheu-Grau, D.; Torres-Odio, S.; Sanchez-Guerrero, A.; Senderovich, O.; Meyron-Holtz, E.G.; Milosevic, I.; Frahm, J.; et al. Impaired lysosomal acidification triggers iron deficiency and inflammation in vivo. *eLife* **2019**, *8*, 51031. [CrossRef]
141. Hughes, C.E.; Coody, T.K.; Jeong, M.-Y.; Berg, J.A.; Winge, D.R.; Hughes, A.L. Cysteine Toxicity Drives Age-Related Mitochondrial Decline by Altering Iron Homeostasis. *Cell* **2020**, *180*, 296–310.e18. [CrossRef]
142. Weber, R.A.; Yen, F.S.; Nicholson, S.P.; Alwaseem, H.; Bayraktar, E.C.; Alam, M.; Timson, R.C.; La, K.; Abu-Remaileh, M.; Molina, H.; et al. Maintaining Iron Homeostasis Is the Key Role of Lysosomal Acidity for Cell Proliferation. *Mol. Cell* **2020**, *77*, 645–655.e7. [CrossRef]
143. Rogers, J.T.; Cahill, C.M. Iron Responsiveness to Lysosomal Disruption: A Novel Pathway to Alzheimer's Disease. *J. Alzheimer's Dis.* **2023**, *96*, 41–45. [CrossRef] [PubMed]
144. Villalón-García, I.; Álvarez-Córdoba, M.; Povea-Cabello, S.; Talaverón-Rey, M.; Villanueva-Paz, M.; Luzón-Hidalgo, R.; Suárez-Rivero, J.M.; Suárez-Carrillo, A.; Munuera-Cabeza, M.; Salas, J.J.; et al. Vitamin E prevents lipid peroxidation and iron accumulation in PLA2G6-Associated Neurodegeneration. *Neurobiol. Dis.* **2022**, *165*, 105649. [CrossRef] [PubMed]
145. Giustizieri, M.; Petrillo, S.; D'Amico, J.; Torda, C.; Quatrana, A.; Vigevano, F.; Specchio, N.; Piemonte, F.; Cherubini, E. The ferroptosis inducer RSL3 triggers in-tericta epileptiform activity in mice cortical neurons. *Front. Cell. Neurosci.* **2023**, *17*, 1213732. [CrossRef] [PubMed]
146. Wu, L.; Xian, X.; Tan, Z.; Dong, F.; Xu, G.; Zhang, M.; Zhang, F. The Role of Iron Metabolism, Lipid Metabolism, and Redox Homeostasis in Alzheimer's Disease: From the Perspective of Ferroptosis. *Mol. Neurobiol.* **2023**, *60*, 2832–2850. [CrossRef]
147. Fassio, A.; Falace, A.; Esposito, A.; Aprile, D.; Guerrini, R.; Benfenati, F. Emerging Role of the Autophagy/Lysosomal Degradative Pathway in Neurodevelopmental Disorders with Epilepsy. *Front. Cell. Neurosci.* **2020**, *14*, 39. [CrossRef]
148. Zhao, W.; Gao, X.; Qiu, S.; Gao, B.; Gao, S.; Zhang, X.; Kang, D.; Han, W.; Dai, P.; Yuan, Y. A subunit of V-ATPases, ATP6V1B2, underlies the pathology of intellectual disability. *EBioMedicine* **2019**, *45*, 408–421. [CrossRef]
149. Colacurcio, D.J.; Nixon, R.A. Disorders of lysosomal acidification-The emerging role of v-ATPase in aging and neu-rodegenerative disease. *Ageing Res. Rev.* **2016**, *32*, 75–88. [CrossRef]

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